

Single-cell RNA-seq Scientific Report

Single-cell RNA-seq Complete Scientific Report · Main scientific narrative · Workflow 6.3.4.3 · BioFlow Local 11.12.8

Delivery model: this Word document is the readable main report. Detailed review tables are in `scrna_supplementary_results.xlsx`; complete lossless CSV/JSON results and provenance are in `scrna_full_results_package.zip`.

Portfolio edition: internal project/session identifiers, dataset record IDs, cryptographic hashes, runtime bundle filenames, local/internal paths, and sample/subject identifiers have been omitted or anonymized. Scientific methods, parameters, figures, numerical results, statistical boundaries, and reproducibility software versions are unchanged.

Research Use Only (RUO). For research and exploratory analysis. Not intended for clinical diagnosis, treatment selection, or patient-management decisions.

1. Executive summary

Measure	Result
Input cells	26306
Retained cells	26306
Retained genes	31114
Biological samples	25
Conditions	B_ALL, T_ALL
Clusters	25
Annotated clusters	25
Unique cell-population labels	8
Composition samples	25
Differential-abundance contrast	T_ALL vs B_ALL
DA significant cell populations (FDR < 0.05)	0
Pseudobulk contrast	T_ALL vs B_ALL
Pseudobulk DE tests	0
Significant pseudobulk DE tests (FDR < 0.05)	0
Exploratory Reactome ORA associations passing current within-population FDR rule	995

Cell populations identified: Cluster 0: T cell (medium confidence); Cluster 1: Unassigned (low confidence); Cluster 2: B lineage · subtype unresolved (medium confidence); Cluster 3: T cell (medium confidence); Cluster 4: Unassigned (low confidence); Cluster 5: Monocyte/macrophage (high confidence); Cluster 6: T cell (high confidence); Cluster 7: Plasma cell (medium confidence); Cluster 8: B cell (high confidence); Cluster 9: Plasma cell (high confidence); Cluster 10: Plasma cell (high confidence); Cluster 11: Plasma cell (high confidence); Cluster 12: T cell (high confidence); Cluster 13: T cell (high confidence); Cluster 14: B lineage · subtype unresolved (high confidence); Cluster 15: Plasma cell (medium confidence); Cluster 16: B lineage · subtype unresolved (high confidence); Cluster 17: B lineage · subtype unresolved (medium confidence); Cluster 18: T/NK lineage · subtype unresolved (high confidence); Cluster 19: B cell (medium confidence); Cluster 20: Platelet (high confidence); Cluster 21: Platelet (high confidence); Cluster 22: B cell (medium confidence); Cluster 23: B cell (medium confidence); Cluster 24: Plasma cell (high confidence)

Pseudobulk DE boundary: Inference is blocked because at least one contrast group has fewer than the required number of independent experimental units. Configured engine: edgeR quasi-likelihood GLM. Annotation-dependent publishability is controlled separately by the annotation qualification gate.

2. Analysis identity and dataset

Measure	Result
Project ID	Omitted in portfolio version
Analysis session ID	Omitted in portfolio version
Dataset	Merged scRNA-seq H5AD dataset
Generated at	2026-09-14T07:50:36.823498+00:00
Workflow	bioflow.scrna.differential_abundance.v6.3.4
Workflow runtime version	6.3.4.3
Requested orientation	auto
Resolved orientation	Cells × Genes (AnnData)
Orientation confidence	intrinsic
Grounding source	persisted_project_session
Recomputed	Yes

Analysis Design overlay

Measure	Result
Schema	bioflow.scrna.analysis_design.v2
Default source column	diagnosis
Target field	analysis_group
Canonical scientific field	condition
Condition assignment unit	sample_id
Unmatched policy	unassigned
Design fingerprint	Omitted in portfolio version
Design rules status	DESIGN RULES VALID
Inference preflight status	INSUFFICIENT INDEPENDENT REPLICATES
Independent replicate counts	{'T_ALL': 2, 'B_ALL': 15}
Total biological samples	25
Included samples	22
Excluded samples	3
Condition counts	{'B_ALL': 18, 'T_ALL': 4}
Unassigned included samples	0
Source metadata unchanged	Yes

Cohort filter

Section	Logic	Order	Column	Operator	Value	Output	Case Sensitive
cohort	AND	1	diagnosis	not_equals	Acute myeloid leukemia	—	No

Derived-condition rules

Section	Logic	Order	Column	Operator	Value	Output	Case Sensitive
rule 1	AND	1	diagnosis	equals	B-cell acute lymphoblastic leukemia	B_ALL	No
rule 2	ELSE	1	—	else	—	T_ALL	No

Analysis Design is an overlay: source metadata are preserved unchanged. Cohort and condition rules are evaluated at the biological-sample level. Cohort-excluded samples receive no canonical condition and therefore do not enter condition-level contrasts; the frozen cell-level QC, clustering, marker, and other Step 1–19 algorithm implementations are unchanged.

3. Biological sample metadata and design

Measure	Result
Matched cells	26306
Coverage (%)	100
Biological samples	25
Metadata scope	cell joined from external metadata
Unmatched metadata rows	0
QC-removed cells with resolved metadata	0
Condition levels	B_ALL, T_ALL
Batch levels	—
Subjects / donors	20
Subjects with multiple samples	5
Subjects spanning conditions	0
Subjects nested within one condition	5
Subjects with multiple timepoints	0
Subjects with multiple tissues	0
Subject ID source	metadata column:subject_id
Raw aggregation unit	sample_id
Effective independent replicate unit	subject_id
Blocking unit	none
Inference status	INSUFFICIENT_INDEPENDENT_REPLICATES
Feasibility reason code	INSUFFICIENT_INDEPENDENT_REPLICATES
Condition×batch diagnostic columns	—
Batch adjustment recommended	No

Experimental unit and contrast feasibility

Measure	Result
Raw aggregation unit	sample_id
Effective independent replicate unit	subject_id
Blocking unit	none
Design structure status	REPEATED SUBJECT NESTED WITHIN CONDITION
Inference status	INSUFFICIENT_INDEPENDENT_REPLICATES
Feasibility reason code	INSUFFICIENT_INDEPENDENT_REPLICATES
Feasibility reason	Inference is blocked because at least one contrast group has fewer than the required number of independent experimental units.
Raw units by group	{'T_ALL': 4, 'B_ALL': 18}
Independent units by group	{'T_ALL': 2, 'B_ALL': 15}
Minimum independent units required per group	3
Automatic subject-level collapse	Yes
Subjects total	17
Subjects with multiple samples	5
Subjects spanning conditions	0
Subjects nested within one condition	5
Subjects with multiple timepoints	0
Subjects with multiple tissues	0

Experimental-unit notices

- Repeated biological samples are nested within one condition per subject; safe inference uses subject-level collapse.

Sample-level QC

Sample Id	Condition	Batch	Subject Id	Cells Input	Cells Retained	Retention Pct	Median Total Counts	Median N Genes	Median Pct Mito
Sample_01	B_ALL	—	Subject_01	1414	1414	100	3589.918154	2333	1.197295
Sample_02	B_ALL	—	Subject_02	75	75	100	5802.018564	3391	1.08577
Sample_03	B_ALL	—	Subject_03	667	667	100	3150.992032	1885	1.347278
Sample_04	T_ALL	—	Subject_04	1676	1676	100	2900.425162	1865.5	1.295669
Sample_05	T_ALL	—	Subject_04	89	89	100	3298.960866	1946	1.194418
Sample_06	T_ALL	—	Subject_05	1985	1985	100	2474.28287	1708	1.705466
Sample_07	T_ALL	—	Subject_05	446	446	100	3688.48164	2300	1.261498
Sample_08	B_ALL	—	Subject_06	827	827	100	2221.83991	1537	1.404856
Sample_09	B_ALL	—	Subject_06	125	125	100	3500.951632	2242	1.172501
Sample_10	—	—	Subject_07	39	39	100	2423.592812	1393	1.767411
Sample_11	B_ALL	—	Subject_08	51	51	100	3679.726979	2294	1.343381
Sample_12	B_ALL	—	Subject_09	241	241	100	3522.403046	2305	1.037355
Sample_13	B_ALL	—	Subject_10	46	46	100	4123.082636	2186.5	1.204179
Sample_14	B_ALL	—	Subject_11	389	389	100	3773.986975	2269	1.422026
Sample_15	B_ALL	—	Subject_12	157	157	100	3023.612665	1830	1.6167
Sample_16	B_ALL	—	Subject_13	1281	1281	100	2117.727701	1405	1.633523
Sample_17	B_ALL	—	Subject_14	2477	2477	100	2519.338004	1686	1.604745
Sample_18	B_ALL	—	Subject_14	1234	1234	100	3656.90591	2345.5	1.367857
Sample_19	B_ALL	—	Subject_15	292	292	100	3854.745716	2338	1.487513
Sample_20	—	—	Subject_16	1884	1884	100	6474.402193	3963	1.035436
Sample_21	B_ALL	—	Subject_17	1141	1141	100	4300.610289	2654	1.1953
Sample_22	B_ALL	—	Subject_18	1677	1677	100	3899.856742	2428	1.340802
Sample_23	B_ALL	—	Subject_19	3114	3114	100	3777.187674	2383	1.388051
Sample_24	B_ALL	—	Subject_19	3712	3712	100	4692.387504	2852.5	1.122152
Sample_25	—	—	Subject_20	1267	1267	100	2468.846665	1214	1.656895

Complete detailed result: [scrna_sample_qc_summary.csv](#) (also indexed in [scRNA Supplementary Results.xlsx](#) and the Full Results Package).

Sample design

Sample Id	Subject Id	Condition	Batch
Sample_01	Subject_01	B_ALL	—
Sample_02	Subject_02	B_ALL	—
Sample_03	Subject_03	B_ALL	—
Sample_04	Subject_04	T_ALL	—
Sample_05	Subject_04	T_ALL	—
Sample_06	Subject_05	T_ALL	—
Sample_07	Subject_05	T_ALL	—
Sample_08	Subject_06	B_ALL	—
Sample_09	Subject_06	B_ALL	—
Sample_10	Subject_07	—	—
Sample_11	Subject_08	B_ALL	—
Sample_12	Subject_09	B_ALL	—
Sample_13	Subject_10	B_ALL	—
Sample_14	Subject_11	B_ALL	—
Sample_15	Subject_12	B_ALL	—
Sample_16	Subject_13	B_ALL	—
Sample_17	Subject_14	B_ALL	—
Sample_18	Subject_14	B_ALL	—

Sample_19	Subject_15	B_ALL	—
Sample_20	Subject_16	—	—
Sample_21	Subject_17	B_ALL	—
Sample_22	Subject_18	B_ALL	—
Sample_23	Subject_19	B_ALL	—
Sample_24	Subject_19	B_ALL	—
Sample_25	Subject_20	—	—

Complete detailed result: scrna_sample_design.csv (also indexed in scRNA Supplementary Results.xlsx and the Full Results Package).

4. Quality control and filtering

Measure	Result
Method	robust median mad filter v1
QC mode requested	auto
QC mode resolved	processed_audit
Cell QC action	flag_only
Input processing state	processed
Input interpretation confidence	high
Auto QC policy	secondary_audit
Input source format	local_h5ad
Filter strategy	capability_driven_secondary_qc_flag_only
Cells before	26306
Cells after	26306
Cells removed	0
Flagged cells	713
Genes before	60319
Genes after	31114
Genes removed	29205
Measure	Result
Mitochondrial QC	Available
Recognized MT genes	37
MT recognition rule	case-insensitive resolved gene-symbol prefix MT-; original feature IDs remain unchanged
MT QC reason	—

QC audit thresholds (flag-only)

Measure	Result
Strategy	distribution_aware_median_mad
Lower Mad	3
Upper Mad	4
Mito Mad	3
Min Genes Per Cell	1
Max Genes Per Cell	5775
Min Counts Per Cell	1
Max Pct Mito	2.247422
Hard Max Pct Mito	30
Min Cells Per Gene	3

QC summary statistics

Phase	Metric	Count	Min	P05	P25	Median	P75	P95	Max	Mean
-------	--------	-------	-----	-----	-----	--------	-----	-----	-----	------

Before	total_counts	26306	224.142567	2054.124966	2574.918708	3582.02148	4308.430216	6179.316618	7715.621254	3595.939573
Before	n_genes_by_counts	26306	202	1026	1677	2228	2893	4285.75	7666	2370.826047
Before	pct_counts_mt	26306	0	0.924402	1.14223	1.330891	1.563249	2.002404	7.158448	1.38035
Phase	Metric	Count	Min	P05	P25	Median	P75	P95	Max	Mean
After	total_counts	26306	224.142567	2054.124966	2574.679971	3581.822302	4307.966216	6178.040123	7714.767047	3595.704882
After	n_genes_by_counts	26306	202	1026	1677	2228	2893	4284.75	7664	2370.596442
After	pct_counts_mt	26306	0	0.924414	1.142317	1.331007	1.5634	2.002404	7.158448	1.380448

Secondary QC flag summary

Measure	Result
Mitochondrial Pct Above Robust Max	546
Detected Genes Above Robust Max	167

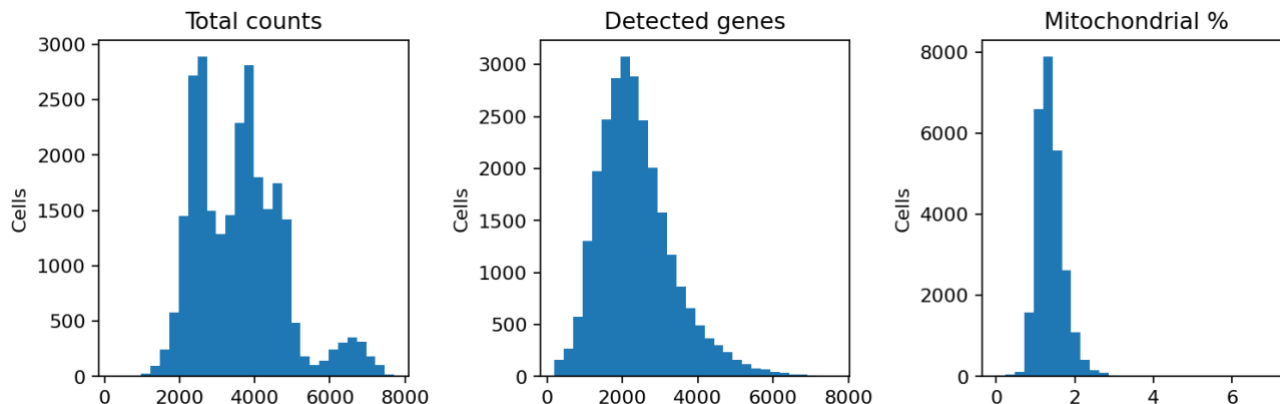


Figure 1. Cell-level QC distributions used for review; complete per-cell values are not expanded in Word.

Per-cell QC metrics and cell/gene filter audits are intentionally not expanded in the main Word report. Complete detailed result: `scrna_qc_cell_metrics.csv`, `scrna_qc_filter_audit.csv`, and `scrna_qc_gene_filter_audit.csv`.

QC warnings

- Upstream-processed H5AD detected: distribution-derived cell QC thresholds are audit flags only. No cells are removed unless the user explicitly selects aggressive re-QC. Gene prevalence filtering remains active for downstream feature efficiency.
- Input capabilities indicate upstream processing, so Auto QC used the existing non-destructive flag-only audit mode. No cells were removed by cell-QC unless aggressive re-QC was explicitly selected.

5. Doublet detection / screening

Measure	Result
Method	qc_high_library_high_complexity_outlier_v1
Action	flag_only
Cells assessed	26306
Candidate cells	257
Removed cells	0

Score threshold	6
-----------------	---

Top candidate cells (up to 20)

Cell Id	Doublet Candidate Score	Doublet Candidate	Action
Cell_001	9.084736	Yes	flag_only
Cell_002	9.032021	Yes	flag_only
Cell_003	9.026444	Yes	flag_only
Cell_004	8.988355	Yes	flag_only
Cell_005	8.924856	Yes	flag_only
Cell_006	8.893796	Yes	flag_only
Cell_007	8.7675	Yes	flag_only
Cell_008	8.595477	Yes	flag_only
Cell_009	8.47359	Yes	flag_only
Cell_010	8.452986	Yes	flag_only
Cell_011	8.37097	Yes	flag_only
Cell_012	8.281456	Yes	flag_only
Cell_013	8.226475	Yes	flag_only
Cell_014	8.200227	Yes	flag_only
Cell_015	8.145667	Yes	flag_only
Cell_016	8.094718	Yes	flag_only
Cell_017	8.09218	Yes	flag_only
Cell_018	8.06725	Yes	flag_only
Cell_019	8.059604	Yes	flag_only
Cell_020	8.015398	Yes	flag_only

Complete detailed result: [scrna_doublet_cell_audit.csv](#) (also indexed in [scRNA Supplementary Results.xlsx](#) and the Full Results Package).

Doublet interpretation notes

- Conservative QC outlier screen only; not Scrublet, DoubletFinder, genotype-aware multiplet detection, or an equivalent probabilistic doublet model.

6. Ambient RNA assessment

Measure	Result
Method	low count tail enrichment risk assessment v1
Correction applied	No
Low-count cells	3946

Top ambient-RNA risk genes (up to 20)

Gene	Gene Symbol	Gene Id	Feature Id	Low Count Mean	Other Cells Mean	Low To Other Ratio	Risk Rank
HBA2	HBA2	ENSG00000188536	ENSG00000188536	2.721648	1.084557	1.785342	1
HBB	HBB	ENSG00000244734	ENSG00000244734	3.233777	1.489498	1.700655	2
HBA1	HBA1	ENSG00000206172	ENSG00000206172	2.209159	0.965877	1.632437	3
HBD	HBD	ENSG00000223609	ENSG00000223609	1.136566	0.370779	1.558651	4
HBM	HBM	ENSG00000206177	ENSG00000206177	0.693269	0.198006	1.413406	5

ALAS2	ALAS2	ENSG00000158578	ENSG00000158578	0.591476	0.153644	1.379521	6
AHSP	AHSP	ENSG00000169877	ENSG00000169877	0.795286	0.305344	1.375336	7
HBG2	HBG2	ENSG00000196565	ENSG00000196565	0.707192	0.274641	1.339351	8
ENSG00000284931	ENSG00000284931	ENSG00000284931	ENSG00000284931	0.427669	0.120148	1.274536	9
SLC25A37	SLC25A37	ENSG00000147454	ENSG00000147454	0.808616	0.420407	1.273308	10
GYPB	GYPB	ENSG00000250361	ENSG00000250361	0.434485	0.135888	1.262875	11
CA1	CA1	ENSG00000133742	ENSG00000133742	0.642752	0.30409	1.259692	12
BPGM	BPGM	ENSG00000172331	ENSG00000172331	0.453186	0.159937	1.252814	13
HBG1	HBG1	ENSG00000213934	ENSG00000213934	0.366868	0.093107	1.250444	14
LUZP2	LUZP2	ENSG00000187398	ENSG00000187398	0.312447	0.050745	1.249063	15
PID1	PID1	ENSG00000153823	ENSG00000153823	0.289334	0.032578	1.248655	16
SNCA	SNCA	ENSG00000145335	ENSG00000145335	0.6154	0.29801	1.24452	17
SELENBP1	SELENBP1	ENSG00000143416	ENSG00000143416	0.339908	0.078013	1.242942	18
SLC4A1	SLC4A1	ENSG00000004939	ENSG00000004939	0.429242	0.150787	1.24197	19
TMCC2	TMCC2	ENSG00000133069	ENSG00000133069	0.262132	0.034875	1.219599	20

Complete detailed result: `scrna_ambient_rna_risk_genes.csv` (also indexed in `scRNA Supplementary Results.xlsx` and the Full Results Package).

Ambient RNA interpretation notes

- Assessment only. Reliable ambient-RNA correction requires raw droplet/empty-droplet evidence and an explicit registered correction method; filtered counts are never presented as SoupX/CellBender-corrected.

7. Normalization

Measure	Result
Method	library size scale then log1p
Target sum	10000
Cells	26306
Genes	31114
Zero-library cells	0
Measure	Result
Input	filtered_counts (raw integer/count scale after QC filtering)
Normalized Counts	library-size scaled counts before log transform
Log1P Expression	log1p(normalized_counts); downstream HVG/PCA/marker foundation input
Raw Counts Preserved	yes; QC/raw matrices are not overwritten by

	normalization
--	---------------

The full per-cell normalization audit is retained in `scrna_normalization_cell_audit.csv` and the Supplementary Results workbook; it is not reproduced cell-by-cell in Word.

8. Highly variable genes

Measure	Result
Method	mean bin normalized dispersion rank v2
Requested HVGs	2000
Genes evaluated	31114
Selected HVGs	2000

Top 30 HVGs

Rank	Gene Symbol	Gene Id	Mean	Dispersion	Normalized Dispersion	Highly Variable
1	HBM	ENSG00000206177	0.167766	2.705207	13.408662	Yes
2	ALAS2	ENSG00000158578	0.157907	2.551643	11.988674	Yes
3	AHSP	ENSG00000169877	0.24759	2.467254	11.398453	Yes
4	SFRP2	ENSG00000145423	0.001035	3.229514	10.809886	Yes
5	IFIT1B	ENSG00000204010	0.037099	2.794383	10.471578	Yes
6	GYPB	ENSG00000250361	0.13233	2.565138	10.369308	Yes
7	CA1	ENSG00000133742	0.250066	2.3336	10.199202	Yes
8	GYPA	ENSG00000170180	0.125051	2.536601	10.137432	Yes
9	RPSAP58	ENSG00000225178	0.000406	3.567215	9.974718	Yes
10	HBD	ENSG00000223609	0.325446	2.265657	9.589563	Yes
11	RPL18AP3	ENSG00000213442	0.004889	2.832727	9.57989	Yes
12	THEM5	ENSG00000196407	0.001745	2.944669	9.392625	Yes
13	SLC4A1	ENSG00000004939	0.146368	2.421718	9.203944	Yes
14	CD7	ENSG00000173762	0.341653	1.947586	8.898067	Yes
15	RPL9P7	ENSG00000238103	0.001368	2.852698	8.835351	Yes
16	RPS3AP26	ENSG00000214389	0.000554	3.082119	8.795628	Yes
17	MYL4	ENSG00000198336	0.124828	2.363693	8.732466	Yes
18	ENSG00000284931	ENSG00000284931	0.120855	2.35726	8.680193	Yes
19	SELENBP1	ENSG00000143416	0.10022	2.437817	8.483312	Yes
20	HBG1	ENSG00000213934	0.096125	2.429517	8.420214	Yes
21	RHAG	ENSG00000112077	0.097549	2.427312	8.403453	Yes
22	RPL21P16	ENSG00000220842	0.068772	2.446004	8.181234	Yes
23	ENSG00000224525	ENSG00000224525	0.000553	2.956662	8.140959	Yes
24	ENSG00000234287	ENSG00000234287	0.000755	2.917213	7.935098	Yes
25	HBQ1	ENSG00000086506	0.082198	2.346435	7.788651	Yes
26	TMCC2	ENSG00000133069	0.059394	2.370158	7.616921	Yes
27	EPB42	ENSG00000166947	0.08113	2.322572	7.607253	Yes
28	ITLN1	ENSG00000179914	0.009995	2.50909	7.602021	Yes
29	HBG2	ENSG00000196565	0.269981	2.043625	7.597313	Yes
30	RUNDC3A	ENSG00000108309	0.009292	2.502822	7.559502	Yes

Complete detailed result: `scrna_hvg.csv` (also indexed in `scRNA Supplementary Results.xlsx` and the Full Results Package).

9. Principal component analysis

Pc	Explained Variance Ratio	Cumulative
----	--------------------------	------------

1	0.076031	0.076031
2	0.063257	0.139288
3	0.050608	0.189897
4	0.026374	0.216271
5	0.024819	0.24109
6	0.016235	0.257325
7	0.015025	0.272349
8	0.012599	0.284948
9	0.012144	0.297093
10	0.008474	0.305567
11	0.00735	0.312917
12	0.006482	0.319399
13	0.006389	0.325788
14	0.005865	0.331653
15	0.005632	0.337285
16	0.0054	0.342685
17	0.005011	0.347696
18	0.004695	0.352391
19	0.004563	0.356954
20	0.004273	0.361226

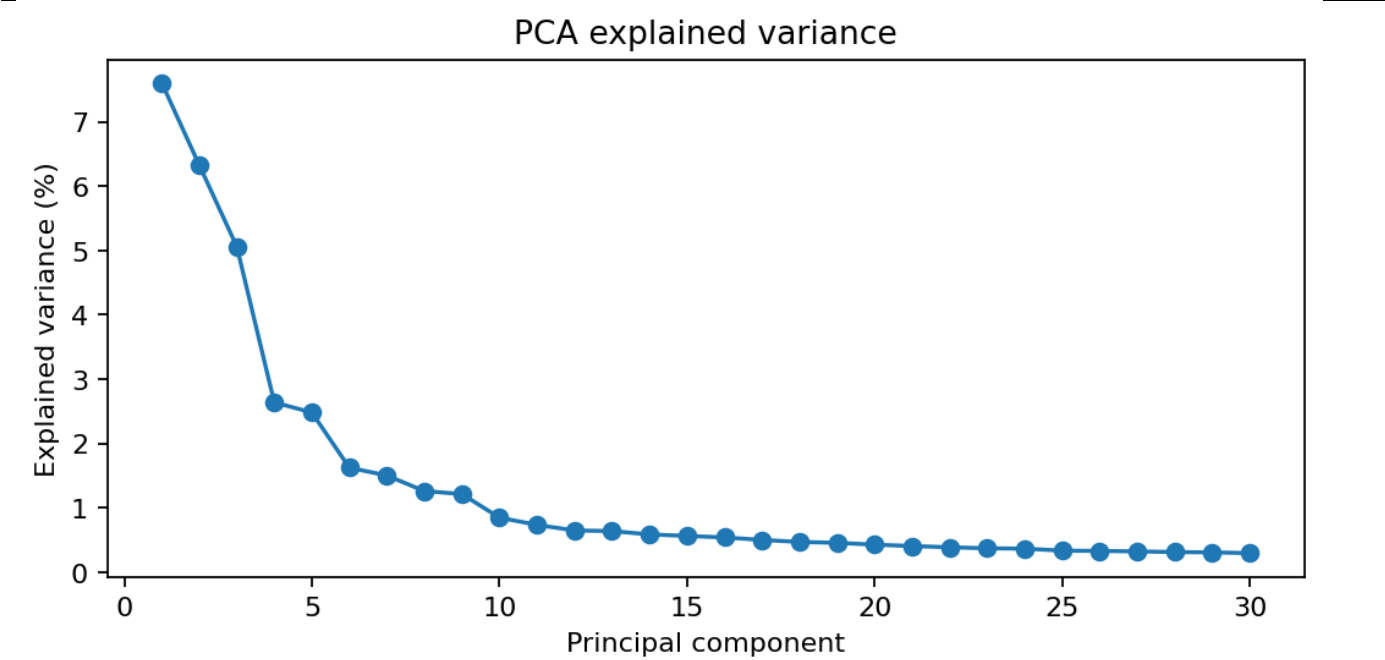


Figure 2. PCA explained variance.

10. Neighbor graph

Measure	Result
Method	sklearn knn euclidean
Neighbors per cell	15
Mean neighbor distance	3.721148
Cells in neighbor index	26306

The neighbor-index matrix is a technical graph object and is not expanded in Word. It remains fully preserved in the step JSON artifact (scrna_step_08_neighbors.json or the corresponding workflow-step artifact).

11. Low-dimensional embedding

Measure	Result
Method	umap-learn
Cells plotted	26306

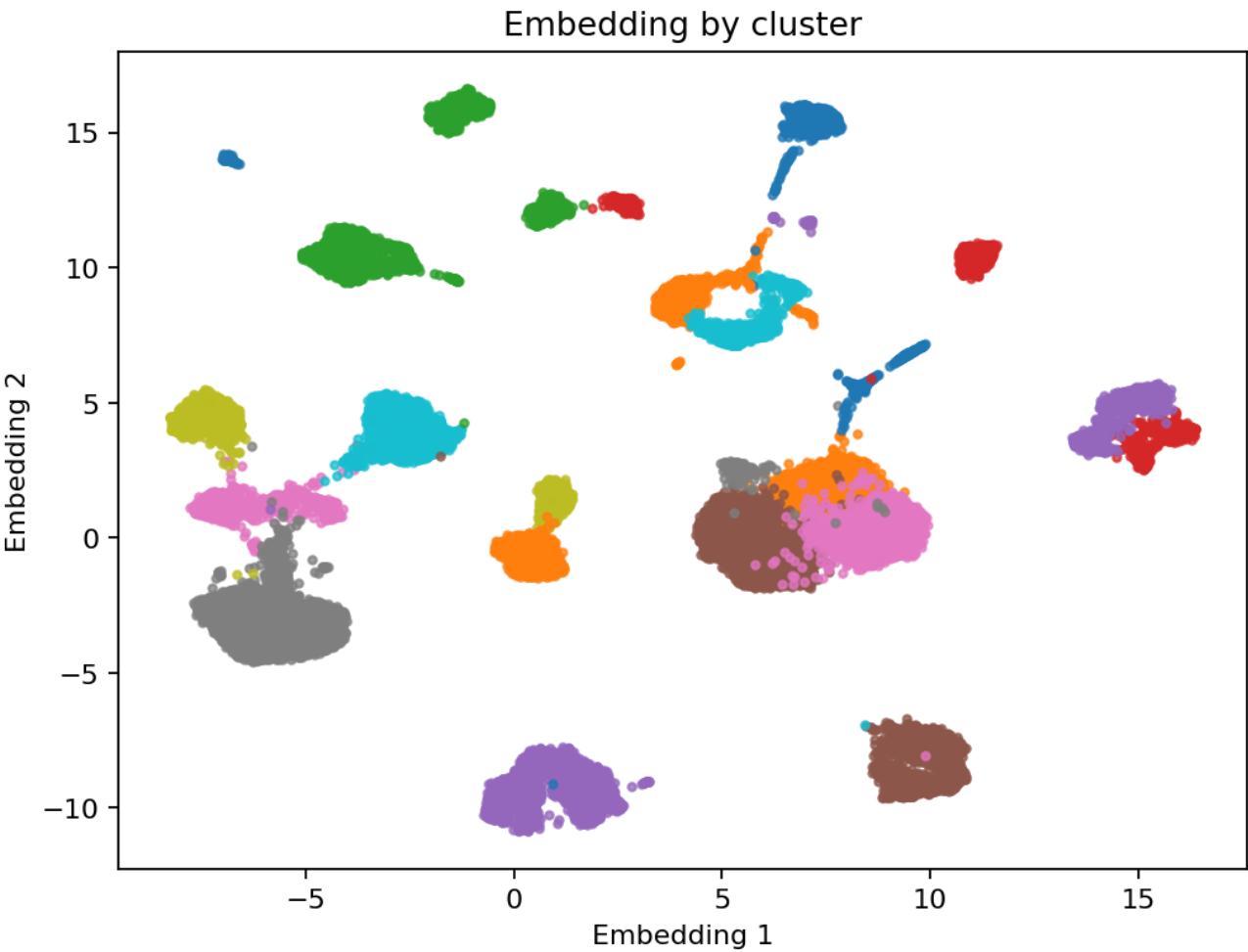


Figure 3. Low-dimensional embedding by cluster.

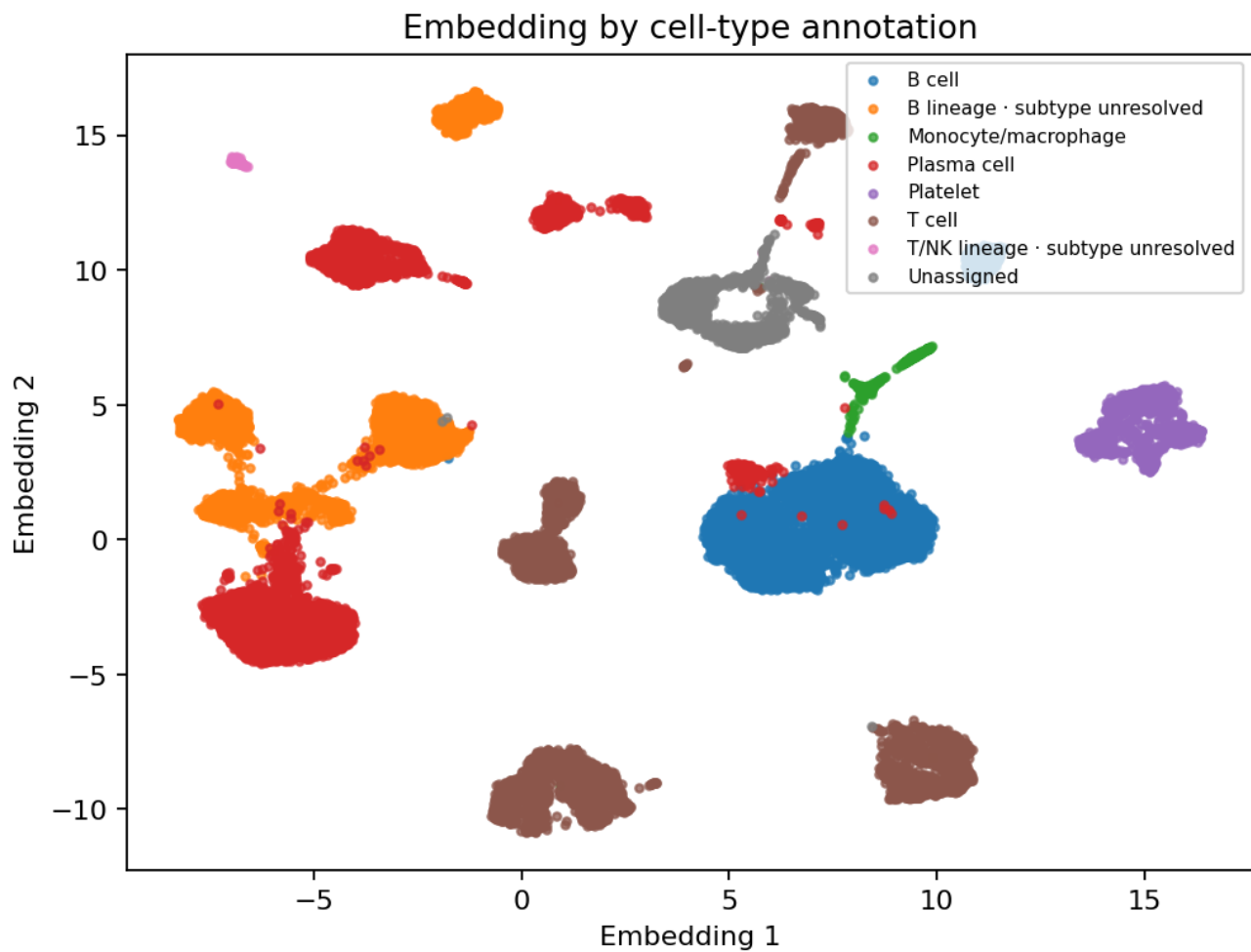


Figure 4. Low-dimensional embedding by cell-type annotation.

Complete embedding coordinates are provided in `scrna_cell_embedding.csv` and are not listed cell-by-cell in Word.

12. Clustering

Measure	Result
Method	Weighted kNN Louvain
Resolution	1
Random seed	63
Fallback used	No
Clusters	25
Graph nodes	26306
Graph edges	294420

Cluster sizes

Cluster	Cell Count
0	709
1	605
2	688
3	414
4	522
5	265

6	790
7	1272
8	415
9	72
10	406
11	220
12	2244
13	1679
14	1105
15	3614
16	1143
17	1558
18	108
19	1763
20	641
21	1104
22	2723
23	2074
24	172

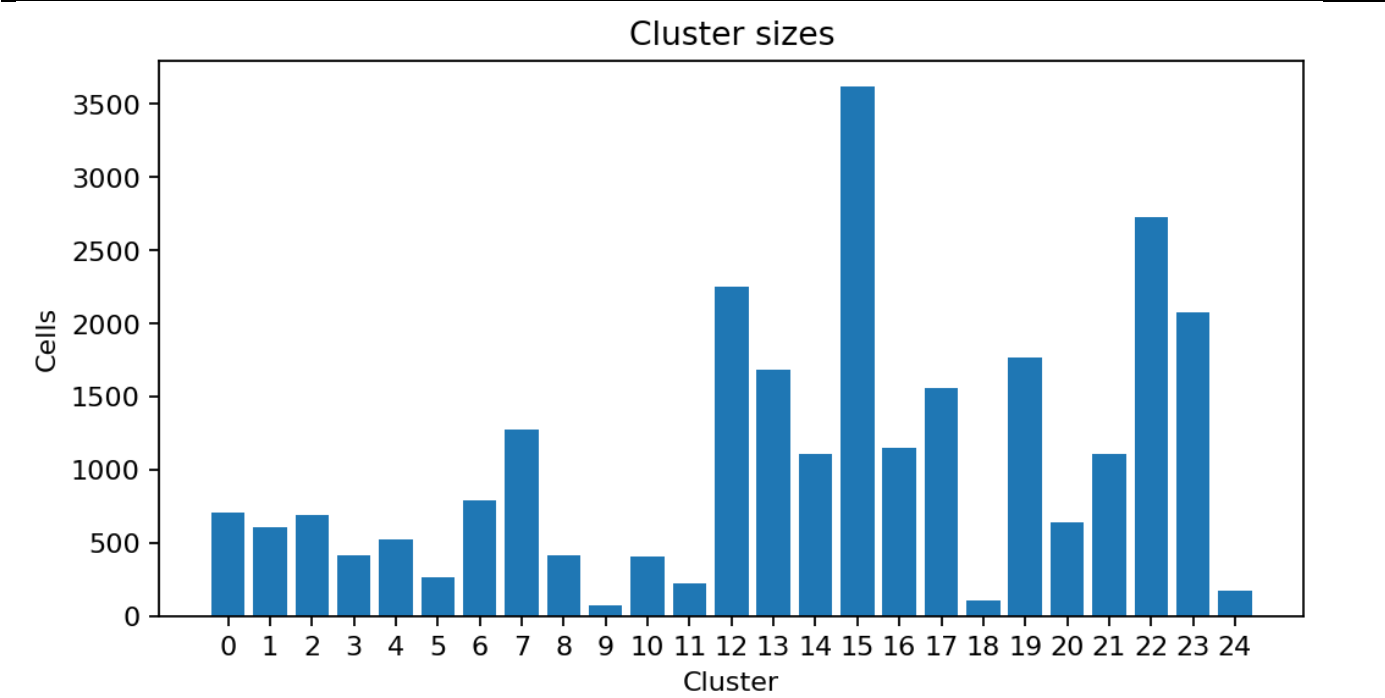


Figure 5. Number of retained cells per cluster.

13. Cluster-vs-rest differential expression

Measure	Result
Method	Cluster-vs-rest Welch + BH FDR
Tests	777850

Top 10 genes per cluster

Cluster	Gene Symbol	Gene Id	Log Expression Delta	Pct Expressin g Cluster	Pct Expressin g Rest	P Value	Fdr
---------	-------------	---------	----------------------	-------------------------	----------------------	---------	-----

0	IGLC3	ENSG00000211679	2.459451	0.992948	0.046451	0	0
0	AGAP1	ENSG00000157985	2.246877	0.981664	0.200375	0	0
0	HBA1	ENSG00000206172	2.218413	1	0.355901	0	0
0	IGLC2	ENSG00000211677	2.032428	0.947814	0.095871	0	0
0	ENSG00000241956	ENSG00000241956	2.006239	0.981664	0.194124	0	0
0	SDK1	ENSG00000146555	1.971796	0.96897	0.170137	0	0
0	IGLL1	ENSG00000128322	1.818035	0.978843	0.270618	0	0
0	ADARB2	ENSG00000185736	1.772016	0.946403	0.204555	0	0
0	NRG3	ENSG00000185737	1.6963	0.923836	0.126499	0	0
0	MT-ND3	ENSG00000198840	-1.695883	0.22567	0.942767	0	0
1	HBD	ENSG00000223609	2.944511	0.993388	0.130034	0	0
1	HBA1	ENSG00000206172	2.791199	1	0.358507	0	0
1	AHSP	ENSG00000169877	2.756601	0.975207	0.082837	0	0
1	HBA2	ENSG00000188536	2.614168	1	0.488541	0	0
1	CA1	ENSG00000133742	2.613878	0.958678	0.08988	0	0
1	BLVRB	ENSG00000090013	2.432656	0.965289	0.16548	0	0
1	PRDX2	ENSG00000167815	2.360079	0.950413	0.152212	0	0
1	HBB	ENSG00000244734	2.319673	1	0.691646	0	0
1	VPREB3	ENSG00000128218	-0.708299	0.028099	0.38232	0	0
1	ENSG00000285947	ENSG00000285947	-0.673373	0.021488	0.398934	0	0
2	AGAP1	ENSG00000157985	1.816766	0.976744	0.201148	0	0
2	ENSG00000241956	ENSG00000241956	1.732791	0.978198	0.194863	0	0
2	PTPRM	ENSG00000173482	1.638948	0.93314	0.136974	0	0
2	PDLIM1	ENSG00000107438	1.267582	0.997093	0.538371	0	0
2	CD24	ENSG00000272398	1.195525	0.992733	0.500234	0	0
2	IGHM	ENSG00000211899	1.105117	0.99564	0.576626	0	0
2	NRXN3	ENSG00000021645	-1.037917	0.049419	0.511125	0	0

2	CXCR4	ENSG00000121966	1.035812	0.981105	0.53115	0	0
2	LST1	ENSG00000204482	-1.009106	0.071221	0.611913	0	0
2	LTB	ENSG00000227507	-0.89699	0.072674	0.507534	0	0
3	CDK14	ENSG00000058091	-0.765719	0.009662	0.459563	0	0
3	GNA15	ENSG00000060558	-0.700736	0.007246	0.46524	0	0
3	SDK2	ENSG00000069188	-0.565493	0.004831	0.362699	0	0
3	ENSG00000285534	ENSG00000285534	-0.467482	0.002415	0.254635	0	0
3	ENSG00000271758	ENSG00000271758	-0.446859	0	0.231886	0	0
3	SH3RF1	ENSG00000154447	-0.444828	0.002415	0.309169	0	0
3	MED12L	ENSG00000144893	-0.411122	0.004831	0.253746	0	0
3	NECTIN2	ENSG00000130202	-0.388069	0	0.277846	0	0
3	DPEP1	ENSG00000015413	-0.385456	0	0.251661	0	0
3	RAB13	ENSG00000143545	-0.36317	0.004831	0.253051	0	0
4	HBD	ENSG00000223609	3.528858	0.998084	0.132718	0	0
4	HBA1	ENSG00000206172	3.436667	1	0.360572	0	0
4	AHSP	ENSG00000169877	3.38666	0.990421	0.085402	0	0
4	HBA2	ENSG00000188536	3.277637	1	0.490188	0	0
4	ALAS2	ENSG00000158578	3.265415	0.998084	0.046385	0	0
4	SLC25A37	ENSG00000147454	3.147601	1	0.230337	0	0
4	CA1	ENSG00000133742	3.10753	0.965517	0.092538	0	0
4	SNCA	ENSG00000145335	2.987373	0.980843	0.160914	0	0
4	BLVRB	ENSG00000090013	2.971825	0.97318	0.167895	0	0
4	HBB	ENSG00000244734	2.933019	1	0.692639	0	0
5	PLEKHG1	ENSG00000120278	-0.446634	0.003774	0.282286	0	0
5	LINC00958	ENSG00000251381	-0.329149	0	0.222226	0	0
5	TCL1B	ENSG00000213231	-0.25783	0	0.118697	0	0
5	ENSG00000259084	ENSG00000259084	-0.2187	0	0.106563	0	0

5	MUC4	ENSG00000145113	-0.212021	0	0.153642	0	0
5	TRDC	ENSG00000211829	-0.192153	0	0.096809	0	0
5	BEST3	ENSG00000127325	-0.191201	0	0.119427	0	0
5	IGF2BP1	ENSG00000159217	-0.185506	0	0.129181	0	0
5	KLRF2	ENSG00000256797	-0.169074	0	0.116739	0	0
5	RAG2	ENSG00000175097	-0.13056	0	0.089551	0	0
6	HLA-DPA1	ENSG00000231389	-1.386708	0.088608	0.715629	0	0
6	EBF1	ENSG00000164330	-1.356622	0.048101	0.69137	0	0
6	DNTT	ENSG00000107447	-1.328717	0.062025	0.706145	0	0
6	MEF2C	ENSG00000081189	-1.307319	0.05443	0.72578	0	0
6	MME	ENSG00000196549	-1.232385	0.021519	0.635288	0	0
6	ERG	ENSG00000157554	-1.223561	0.025316	0.710103	0	0
6	TCF4	ENSG00000196628	-1.158142	0.04557	0.657783	0	0
6	MZB1	ENSG00000170476	-1.151688	0.032911	0.659116	0	0
6	HLA-DRB1	ENSG00000196126	-1.059694	0.050633	0.59061	0	0
6	NRXN3	ENSG00000021645	-1.017799	0.048101	0.513011	0	0
7	ENSG00000287092	ENSG00000287092	1.529445	0.987421	0.382799	0	0
7	PLEKHG1	ENSG00000120278	1.371037	0.915094	0.247184	0	0
7	SNX9	ENSG00000130340	1.291018	0.904874	0.195814	0	0
7	GAS7	ENSG00000007237	1.252841	0.940252	0.339818	0	0
7	IL1B	ENSG00000125538	1.237711	0.793239	0.1029	0	0
7	VPREB1	ENSG00000169575	1.237441	0.990566	0.484661	0	0
7	PDLIM1	ENSG00000107438	1.222522	0.986635	0.528202	0	0
7	PDE4B	ENSG00000184588	1.222474	0.913522	0.312855	0	0
7	CD9	ENSG00000010278	1.22245	0.870283	0.175322	0	0
7	VPREB3	ENSG00000128218	1.174713	0.922956	0.346289	0	0
8	CD74	ENSG00000019582	0.999394	1	0.810513	0	0

8	TCF7L2	ENSG0000014873 7	-0.689655	0.007229	0.452783	0	0
8	PARD3	ENSG0000014849 8	-0.62458	0.007229	0.406087	0	0
8	H1-0	ENSG0000018906 0	-0.574706	0.00241	0.364026	0	0
8	NAV1	ENSG0000013436 9	-0.560326	0.007229	0.371905	0	0
8	SMAD1	ENSG0000017036 5	-0.457119	0.00241	0.304507	0	0
8	ECM1	ENSG0000014336 9	-0.456729	0.00241	0.314472	0	0
8	ENSG0000027175 8	ENSG0000027175 8	-0.443696	0.00241	0.231857	0	0
8	NECTIN2	ENSG0000013020 2	-0.385165	0.00241	0.277819	0	0
8	CTDSPL	ENSG0000014467 7	-0.379152	0	0.2441	0	0
9	P2RY14	ENSG0000017494 4	-0.316903	0	0.214836	0	0
9	TRGC2	ENSG0000022719 1	-0.301079	0	0.190021	0	0
9	ANXA5	ENSG0000016411 1	-0.300925	0	0.210185	0	0
9	BEX3	ENSG0000016668 1	-0.294507	0	0.174125	0	0
9	PLXDC2	ENSG0000012059 4	-0.267945	0	0.167759	0	0
9	IL32	ENSG0000000851 7	-0.266349	0	0.133338	0	0
9	TAFA1	ENSG0000018366 2	-0.258919	0	0.137951	0	0
9	TCL1B	ENSG0000021323 1	-0.255933	0	0.117824	0	0
9	SHROOM3	ENSG0000013877 1	-0.248077	0	0.152474	0	0
9	FLNA	ENSG0000019692 4	-0.24714	0	0.177137	0	0
10	RPL9	ENSG0000016368 2	1.188601	1	0.67471	0	0
10	XIST	ENSG0000022980 7	-0.64286	0	0.393398	0	0
10	S100A10	ENSG0000019774 7	-0.574309	0.004926	0.328919	0	0
10	PCCA	ENSG0000017519 8	-0.512646	0.002463	0.322085	0	0
10	NDFIP1	ENSG0000013150 7	-0.372181	0.002463	0.247027	0	0
10	TFPI	ENSG0000000343 6	-0.313731	0	0.220347	0	0
10	TRGC2	ENSG0000022719 1	-0.304962	0	0.192471	0	0
10	ANXA5	ENSG0000016411 1	-0.301643	0.002463	0.212857	0	0

10	BEX3	ENSG0000016668 1	-0.298305	0	0.176371	0	0
10	TRBC1	ENSG0000021175 1	-0.297349	0.002463	0.142664	0	0
11	RPL9	ENSG0000016368 2	1.177646	1	0.67703	0	0
11	CD44	ENSG0000002650 8	-0.789668	0.004545	0.449666	0	0
11	YES1	ENSG0000017610 5	-0.645637	0	0.415625	0	0
11	S100A10	ENSG0000019774 7	-0.572612	0.004545	0.326612	0	0
11	CTHRC1	ENSG0000016493 2	-0.366944	0	0.250134	0	0
11	BMPR1B	ENSG0000013869 6	-0.355111	0	0.226405	0	0
11	ATXN1	ENSG0000012478 8	-0.343025	0	0.2302	0	0
11	AGPAT4	ENSG0000002665 2	-0.314422	0	0.221153	0	0
11	CD96	ENSG0000015328 3	-0.311952	0	0.201602	0	0
11	RIN3	ENSG0000010059 9	-0.310967	0	0.224373	0	0
12	CD7	ENSG0000017376 2	2.218486	0.971925	0.080542	0	0
12	CD3D	ENSG0000016728 6	2.066111	0.982175	0.181739	0	0
12	DNAJC1	ENSG0000013677 0	1.743675	0.958111	0.357285	0	0
12	RNF220	ENSG0000018714 7	1.722846	0.97549	0.360111	0	0
12	CD74	ENSG0000001958 2	-1.688064	0.325758	0.858989	0	0
12	HLA-DRA	ENSG0000020428 7	-1.659946	0.258021	0.818843	0	0
12	TRBC2	ENSG0000021177 2	1.628945	0.900178	0.183692	0	0
12	TRDC	ENSG0000021182 9	1.623469	0.804813	0.029715	0	0
12	HLA-DPB1	ENSG0000022386 5	-1.570073	0.121212	0.777284	0	0
12	S100A10	ENSG0000019774 7	1.542331	0.911319	0.269138	0	0
13	TRBC1	ENSG0000021175 1	2.235197	0.947588	0.085475	0	0
13	CD3D	ENSG0000016728 6	1.846384	0.94461	0.202664	0	0
13	CD74	ENSG0000001958 2	-1.643207	0.342466	0.845617	0	0
13	HLA-DRA	ENSG0000020428 7	-1.592752	0.274568	0.804848	0	0
13	HLA-DPB1	ENSG0000022386 5	-1.549452	0.114949	0.762659	0	0

13	CASC15	ENSG00000272168	1.537845	0.979154	0.425427	0	0
13	AUTS2	ENSG00000158321	-1.49736	0.089934	0.817517	0	0
13	MAL	ENSG00000172005	1.44992	0.782013	0.056483	0	0
13	HLA-DPA1	ENSG00000231389	-1.428778	0.114354	0.736509	0	0
13	EBF1	ENSG00000164330	-1.416946	0.05539	0.714094	0	0
14	IFITM1	ENSG00000185885	-0.823204	0.040724	0.459744	0	0
14	PLCB1	ENSG00000182621	-0.703328	0.072398	0.468394	0	0
14	RPS4Y1	ENSG00000129824	-0.684848	0.065158	0.408912	0	0
14	CD74	ENSG00000019582	0.660103	0.99457	0.805563	0	0
14	JCHAIN	ENSG00000132465	-0.649764	0.035294	0.342566	0	0
14	CLEC11A	ENSG00000105472	-0.475812	0.011765	0.326495	0	0
14	PRSS57	ENSG00000185198	-0.459152	0.01629	0.299314	0	0
14	TRGC2	ENSG00000227191	-0.298015	0.011765	0.197294	0	0
14	BEX3	ENSG00000166681	-0.296704	0.00724	0.180945	0	0
14	PSTPIP2	ENSG00000152229	-0.267502	0.011765	0.204992	0	0
15	ENSG00000260371	ENSG00000260371	2.0081	0.959602	0.217213	0	0
15	RPS4Y2	ENSG00000280969	1.86301	0.924737	0.05253	0	0
15	RPL10P9	ENSG00000233913	1.749195	0.913669	0.186982	0	0
15	TCL1B	ENSG00000213231	1.743447	0.798561	0.009034	0	0
15	STAG3	ENSG00000066923	1.661438	0.909795	0.188525	0	0
15	ARPP21	ENSG00000172995	1.472932	0.885169	0.248414	0	0
15	ENSG00000227706	ENSG00000227706	1.466779	0.882402	0.237396	0	0
15	ENSG00000259084	ENSG00000259084	1.398711	0.669618	0.015644	0	0
15	CD79A	ENSG00000105369	1.373332	0.935529	0.459016	0	0
15	GNG7	ENSG00000176533	1.348525	0.952407	0.439318	0	0
16	HLA-DRA	ENSG00000204287	1.147663	0.9965	0.76076	0	0
16	HLA-DPB1	ENSG00000223865	0.998031	0.967629	0.71013	0	0

16	CD74	ENSG00000019582	0.968972	0.9965	0.80519	0	0
16	IFITM1	ENSG00000185885	-0.851668	0.021872	0.461233	0	0
16	CSTB	ENSG00000160213	-0.777073	0.041995	0.517625	0	0
16	SRGN	ENSG00000122862	-0.678005	0.040245	0.453205	0	0
16	CCDC26	ENSG00000229140	-0.653951	0.055118	0.396654	0	0
16	XIST	ENSG00000229807	-0.645134	0.009624	0.404483	0	0
16	CD34	ENSG00000174059	-0.58671	0.016623	0.365298	0	0
16	ENSG00000288681	ENSG00000288681	-0.560689	0.03937	0.401542	0	0
17	HS3ST4	ENSG00000182601	1.709605	0.865854	0.030427	0	0
17	ENSG00000285534	ENSG00000285534	1.607702	0.963415	0.205794	0	0
17	ENSG00000237550	ENSG00000237550	-1.306917	0.525032	0.915306	0	0
17	MDK	ENSG00000110492	1.210603	0.895379	0.321683	0	0
17	ENSG00000271758	ENSG00000271758	1.173599	0.799743	0.192258	0	0
17	APOO	ENSG00000184831	-1.162353	0.462131	0.829724	0	0
17	NRN1	ENSG00000124785	1.112	0.760591	0.122959	0	0
17	TCL1A	ENSG00000100721	1.07727	0.955712	0.505293	0	0
17	VPREB1	ENSG00000169575	-1.007235	0.040436	0.538629	0	0
17	RPL9	ENSG00000163682	0.993089	0.996791	0.65977	0	0
18	JCHAIN	ENSG00000132465	-0.682598	0	0.331018	0	0
18	CD34	ENSG00000174059	-0.593418	0	0.351592	0	0
18	PAM	ENSG00000145730	-0.414426	0	0.27754	0	0
18	SUSD3	ENSG00000157303	-0.37434	0	0.263455	0	0
18	C12orf75	ENSG00000235162	-0.372354	0	0.255287	0	0
18	PECAM1	ENSG00000261371	-0.3224	0	0.234293	0	0
18	BTBD11	ENSG00000151136	-0.319925	0	0.200168	0	0
18	BIN2	ENSG00000110934	-0.307582	0	0.206352	0	0
18	TRAC	ENSG00000277734	-0.299328	0	0.189862	0	0

18	GPRIN3	ENSG0000018547 7	-0.275379	0	0.18196	0	0
19	JCHAIN	ENSG0000013246 5	1.489068	0.958026	0.284521	0	0
19	NRXN3	ENSG0000002164 5	1.484617	0.982416	0.464328	0	0
19	CRADD	ENSG0000016937 2	1.406597	0.972206	0.351913	0	0
19	ID1	ENSG0000012596 8	1.36275	0.922859	0.338997	0	0
19	ID3	ENSG0000011731 8	1.275162	0.972774	0.463635	0	0
19	S100A16	ENSG0000018864 3	1.23019	0.966534	0.359288	0	0
19	RPL36A	ENSG0000024134 3	-1.10025	0.013046	0.507151	0	0
19	CCDC26	ENSG0000022914 0	1.047646	0.862167	0.347309	0	0
19	RPL17-C18orf32	ENSG0000021547 2	1.041623	0.996029	0.504584	0	0
19	CD34	ENSG0000017405 9	1.037241	0.880885	0.312024	0	0
20	MTND1P23	ENSG0000022597 2	1.986608	0.99532	0.074654	0	0
20	CD74	ENSG0000001958 2	-1.742806	0.347894	0.825132	0	0
20	HLA-DRA	ENSG0000020428 7	-1.715544	0.24025	0.784259	0	0
20	CD52	ENSG0000016944 2	-1.629186	0.120125	0.796221	0	0
20	MTCO1P12	ENSG0000023797 3	1.552166	0.970359	0.051276	0	0
20	RPL10P9	ENSG0000023391 3	1.542181	1	0.269004	0	0
20	WASH6P	ENSG0000018248 4	1.524067	0.940718	0.086538	0	0
20	HOPX	ENSG0000017147 6	1.514714	0.971919	0.113384	0	0
20	FCER1G	ENSG0000015886 9	1.500497	0.953198	0.065732	0	0
20	PRSS57	ENSG0000018519 8	1.465144	0.99376	0.269784	0	0
21	MTND1P23	ENSG0000022597 2	1.972847	0.990942	0.057932	0	0
21	CD74	ENSG0000001958 2	-1.764616	0.391304	0.831997	0	0
21	HLA-DRA	ENSG0000020428 7	-1.743459	0.259058	0.793429	0	0
21	CD52	ENSG0000016944 2	-1.637566	0.151268	0.807277	0	0
21	WASH6P	ENSG0000018248 4	1.583862	0.964674	0.069796	0	0
21	MTCO1P12	ENSG0000023797 3	1.539834	0.967391	0.034521	0	0

21	FCER1G	ENSG00000158869	1.526857	0.961051	0.049083	0	0
21	LINGO2	ENSG00000174482	1.524661	0.935688	0.051702	0	0
21	RPL10P9	ENSG00000233913	1.518504	1	0.255575	0	0
21	HOPX	ENSG00000171476	1.502142	0.977355	0.097373	0	0
22	NRXN3	ENSG00000021645	1.506439	0.985678	0.442861	0	0
22	JCHAIN	ENSG00000132465	1.448449	0.938671	0.259339	0	0
22	CRADD	ENSG00000169372	1.443066	0.973926	0.326464	0	0
22	ID1	ENSG00000125968	1.372879	0.918472	0.315736	0	0
22	ID3	ENSG00000117318	1.299638	0.969152	0.443328	0	0
22	S100A16	ENSG00000188643	1.262899	0.968417	0.334351	0	0
22	RPL36A	ENSG00000241343	1.239233	0.995226	0.413857	0	0
22	CD34	ENSG00000174059	1.151285	0.914433	0.284993	0	0
22	CCDC26	ENSG00000229140	1.068786	0.869996	0.325446	0	0
22	HLA-DQA1	ENSG00000196735	1.054722	0.945648	0.450536	0	0
23	NRXN3	ENSG00000021645	1.52858	0.988428	0.457164	0	0
23	JCHAIN	ENSG00000132465	1.524302	0.956606	0.275999	0	0
23	CRADD	ENSG00000169372	1.448371	0.979749	0.343306	0	0
23	ID1	ENSG00000125968	1.380728	0.922854	0.331504	0	0
23	ID3	ENSG00000117318	1.281516	0.96866	0.457453	0	0
23	RPL17-C18orf32	ENSG00000215472	-1.276284	0.045805	0.579605	0	0
23	S100A16	ENSG00000188643	1.253889	0.965767	0.35156	0	0
23	RPL36A	ENSG00000241343	-1.091833	0.027483	0.512257	0	0
23	CD34	ENSG00000174059	1.066007	0.878978	0.304886	0	0
23	CCDC26	ENSG00000229140	1.059113	0.862584	0.340665	0	0
24	FAM30A	ENSG00000226777	-0.131911	0	0.087931	0	0
24	ENSG00000254951	ENSG00000254951	-0.125708	0	0.075457	0	0
24	APBA2	ENSG00000034053	-0.11584	0	0.079169	0	0

24	TAF5	ENSG00000219438	-0.111634	0	0.072511	0	0
24	NOS1AP	ENSG00000198929	-0.10199	0	0.078557	0	0
24	MPPED2	ENSG00000066382	-0.092643	0	0.066924	0	0
24	DOK6	ENSG00000206052	-0.088026	0	0.064246	0	0
24	CYB5R2	ENSG00000166394	-0.07703	0	0.054718	1.16092e-303	4.51509e-300
24	ENSG00000284418	ENSG00000284418	-0.07345	0	0.054488	1.81923e-300	6.2893e-297
24	PCLO	ENSG00000186472	-0.201586	0.005814	0.120533	4.09476e-295	1.27404e-291

Complete detailed result: [scrna_cluster_differential_expression.csv](#) (also indexed in [scRNA Supplementary Results.xlsx](#) and the Full Results Package).

Cluster-DE interpretation notes

- Cluster-vs-rest cell-level testing is used as exploratory marker evidence. It is not biological-replicate condition inference; condition comparisons use sample-level pseudobulk models.

14. Marker genes

Measure	Result
Method	de evidence ranked markers v1
Configured top N per cluster	25
Exploratory fallback clusters	—

Top marker genes per cluster (up to 10)

Cluster	Rank	Gene Symbol	Gene Id	Log Expression Delta	Pct Expressing Cluster	Fdr
0	1	IGLC3	ENSG00000211679	2.459451	0.992948	0
0	2	AGAP1	ENSG00000157985	2.246877	0.981664	0
0	3	HBA1	ENSG00000206172	2.218413	1	0
0	4	IGLC2	ENSG00000211677	2.032428	0.947814	0
0	5	ENSG00000241956	ENSG00000241956	2.006239	0.981664	0
0	6	SDK1	ENSG00000146555	1.971796	0.96897	0
0	7	IGLL1	ENSG00000128322	1.818035	0.978843	0
0	8	ADARB2	ENSG00000185736	1.772016	0.946403	0
0	9	NRG3	ENSG00000185737	1.6963	0.923836	0
0	10	PDLIM1	ENSG00000107438	1.688194	0.99859	0
1	1	HBD	ENSG00000223609	2.944511	0.993388	0
1	2	HBA1	ENSG00000206172	2.791199	1	0
1	3	AHSP	ENSG00000169877	2.756601	0.975207	0
1	4	HBA2	ENSG00000188536	2.614168	1	0
1	5	CA1	ENSG00000133742	2.613878	0.958678	0
1	6	BLVRB	ENSG00000090013	2.432656	0.965289	0
1	7	PRDX2	ENSG00000167815	2.360079	0.950413	0
1	8	HBB	ENSG00000244734	2.319673	1	0
1	9	GYPB	ENSG00000250361	2.270275	0.943802	2.27641e-314

1	10	SNCA	ENSG00000145335	2.027183	0.947107	4.05018e-309
2	1	AGAP1	ENSG00000157985	1.816766	0.976744	0
2	2	ENSG00000241956	ENSG00000241956	1.732791	0.978198	0
2	3	PTPRM	ENSG00000173482	1.638948	0.93314	0
2	4	PDLIM1	ENSG00000107438	1.267582	0.997093	0
2	5	CD24	ENSG00000272398	1.195525	0.992733	0
2	6	IGHM	ENSG00000211899	1.105117	0.99564	0
2	7	CXCR4	ENSG00000121966	1.035812	0.981105	0
2	8	TCF4	ENSG00000196628	0.865375	0.991279	0
2	9	EBF1	ENSG00000164330	0.786752	0.991279	0
2	10	RCSD1	ENSG00000198771	0.778287	0.997093	0
3	1	NKG7	ENSG00000105374	2.605299	0.971014	2.36509e-284
3	2	B2M	ENSG00000166710	0.620315	1	2.89134e-244
3	3	TMSB4X	ENSG00000205542	0.54854	1	4.24005e-231
3	4	CCL5	ENSG00000271503	2.506185	0.913043	5.78216e-205
3	5	HLA-B	ENSG00000234745	0.729872	0.995169	2.16646e-196
3	6	RPS27	ENSG00000177954	0.451454	1	2.44403e-163
3	7	CST7	ENSG00000077984	2.053519	0.855072	1.59742e-160
3	8	MALAT1	ENSG00000251562	0.40259	1	2.89927e-153
3	9	HCST	ENSG00000126264	1.615904	0.896135	1.97779e-143
3	10	RPS27A	ENSG00000143947	0.335294	1	3.44598e-131
4	1	HBD	ENSG00000223609	3.528858	0.998084	0
4	2	HBA1	ENSG00000206172	3.436667	1	0
4	3	AHSP	ENSG00000169877	3.38666	0.990421	0
4	4	HBA2	ENSG00000188536	3.277637	1	0
4	5	ALAS2	ENSG00000158578	3.265415	0.998084	0
4	6	SLC25A37	ENSG00000147454	3.147601	1	0
4	7	CA1	ENSG00000133742	3.10753	0.965517	0
4	8	SNCA	ENSG00000145335	2.987373	0.980843	0
4	9	BLVRB	ENSG00000090013	2.971825	0.97318	0
4	10	HBB	ENSG00000244734	2.933019	1	0
5	1	SRGN	ENSG00000122862	1.392992	0.901887	3.36668e-78
5	2	FTL	ENSG00000087086	0.657835	0.992453	6.64056e-67
5	3	TYROBP	ENSG00000011600	1.73358	0.728302	1.65803e-65
5	4	S100A4	ENSG00000196154	1.28958	0.871698	5.64866e-59
5	5	CST3	ENSG00000101439	1.40787	0.709434	3.24159e-52
5	6	LGALS1	ENSG00000100097	1.384969	0.758491	1.22452e-51
5	7	CFD	ENSG00000197766	1.371286	0.630189	3.11115e-51
5	8	VIM	ENSG00000026025	0.821049	0.939623	4.01568e-50
5	9	S100A6	ENSG00000197956	1.163448	0.837736	6.97694e-47
5	10	TSPO	ENSG00000100300	0.962841	0.826415	3.568e-44
6	1	TMSB4X	ENSG00000205542	0.393799	1	6.50796e-

						316
6	2	RPS18	ENSG00000231500	0.452555	0.998734	7.29376e-295
6	3	RPLP1	ENSG00000137818	0.390096	1	2.59997e-288
6	4	RPS27	ENSG00000177954	0.531296	0.997468	5.22383e-282
6	5	RPL28	ENSG00000108107	0.403859	0.998734	6.58548e-276
6	6	RPL41	ENSG00000229117	0.411458	0.998734	2.70922e-266
6	7	RPL37A	ENSG00000197756	0.425716	0.998734	4.61243e-262
6	8	RPL13	ENSG00000167526	0.454311	0.997468	2.66478e-261
6	9	RPL30	ENSG00000156482	0.491072	0.996203	2.00856e-258
6	10	RPLP2	ENSG00000177600	0.462792	0.997468	1.52518e-256
7	1	ENSG00000287092	ENSG00000287092	1.529445	0.987421	0
7	2	PLEKHG1	ENSG00000120278	1.371037	0.915094	0
7	3	SNX9	ENSG00000130340	1.291018	0.904874	0
7	4	GAS7	ENSG00000007237	1.252841	0.940252	0
7	5	IL1B	ENSG00000125538	1.237711	0.793239	0
7	6	VPREB1	ENSG00000169575	1.237441	0.990566	0
7	7	PDLIM1	ENSG00000107438	1.222522	0.986635	0
7	8	PDE4B	ENSG00000184588	1.222474	0.913522	0
7	9	CD9	ENSG00000010278	1.22245	0.870283	0
7	10	VPREB3	ENSG00000128218	1.174713	0.922956	0
8	1	CD74	ENSG00000019582	0.999394	1	0
8	2	CD37	ENSG00000104894	1.558757	0.971084	5.1058e-204
8	3	MS4A1	ENSG00000156738	1.757965	0.838554	1.74661e-136
8	4	HLA-DRA	ENSG00000204287	0.886136	0.975904	3.6028e-124
8	5	RPL13	ENSG00000167526	0.347786	1	1.98855e-122
8	6	RPS25	ENSG00000118181	0.356225	1	8.26887e-121
8	7	RPL34	ENSG00000109475	0.383682	1	3.23205e-120
8	8	RPLP2	ENSG00000177600	0.363015	1	6.82726e-120
8	9	RPL32	ENSG00000144713	0.352335	1	1.82086e-118
8	10	RPL23A	ENSG00000198242	0.394824	0.99759	5.75966e-115
9	1	JCHAIN	ENSG00000132465	1.684268	0.986111	2.53624e-51
9	2	HLA-DRB1	ENSG00000196126	0.886719	1	4.4719e-50
9	3	IFITM1	ENSG00000185885	1.463907	0.986111	3.309e-49
9	4	ENSG00000286848	ENSG00000286848	1.693325	0.972222	9.85667e-47
9	5	HLA-DPB1	ENSG00000223865	0.682563	1	1.39583e-46
9	6	RPS4Y2	ENSG00000280969	1.412137	0.986111	3.02045e-45
9	7	HLA-DPA1	ENSG00000231389	0.667022	1	3.06326e-45

9	8	HLA-DRA	ENSG00000204287	0.542312	1	3.20685e-40
9	9	ARHGAP24	ENSG00000138639	1.258817	0.986111	5.5749e-40
9	10	DNTT	ENSG00000107447	0.733137	1	6.19747e-39
10	1	RPL9	ENSG00000163682	1.188601	1	0
10	2	JCHAIN	ENSG00000132465	1.93266	0.982759	5.04875e-273
10	3	UBASH3B	ENSG00000154127	1.312681	0.987685	1.77757e-231
10	4	CD99	ENSG00000002586	0.843973	0.995074	3.22625e-227
10	5	ENSG00000287092	ENSG00000287092	1.53021	0.980296	4.13976e-217
10	6	DNTT	ENSG00000107447	0.940009	0.992611	1.01952e-201
10	7	RPL10P9	ENSG00000233913	1.596206	0.948276	1.16787e-193
10	8	ADGRE5	ENSG00000123146	1.059729	0.980296	7.56604e-160
10	9	RPS4Y2	ENSG00000280969	1.488292	0.899015	2.89521e-152
10	10	ENSG00000285106	ENSG00000285106	1.307728	0.94335	3.92682e-140
11	1	RPL9	ENSG00000163682	1.177646	1	0
11	2	LAPTM5	ENSG00000162511	0.79943	1	3.78163e-198
11	3	CD9	ENSG00000010278	2.192529	0.995455	6.05194e-198
11	4	CD74	ENSG00000019582	0.708297	1	6.6228e-194
11	5	IGHM	ENSG00000211899	1.30837	0.995455	1.05592e-169
11	6	JMJD1C	ENSG00000171988	0.861449	1	9.3104e-137
11	7	HLA-DRA	ENSG00000204287	0.623984	1	2.15744e-128
11	8	RPL10P9	ENSG00000233913	1.566544	0.972727	3.04144e-128
11	9	RPS4Y2	ENSG00000280969	1.733617	0.968182	4.43183e-128
11	10	VPREB3	ENSG00000128218	1.560566	0.968182	2.522e-118
12	1	CD7	ENSG00000173762	2.218486	0.971925	0
12	2	CD3D	ENSG00000167286	2.066111	0.982175	0
12	3	DNAJC1	ENSG00000136770	1.743675	0.958111	0
12	4	RNF220	ENSG00000187147	1.722846	0.97549	0
12	5	TRBC2	ENSG00000211772	1.628945	0.900178	0
12	6	TRDC	ENSG00000211829	1.623469	0.804813	0
12	7	S100A10	ENSG00000197747	1.542331	0.911319	0
12	8	IFITM1	ENSG00000185885	1.4091	0.909091	0
12	9	IL32	ENSG00000008517	1.40096	0.750446	0
12	10	CD3E	ENSG00000198851	1.393445	0.815954	0
13	1	TRBC1	ENSG00000211751	2.235197	0.947588	0
13	2	CD3D	ENSG00000167286	1.846384	0.94461	0
13	3	CASC15	ENSG00000272168	1.537845	0.979154	0
13	4	MAL	ENSG00000172005	1.44992	0.782013	0
13	5	CHI3L2	ENSG00000064886	1.401153	0.718285	0

13	6	LCK	ENSG00000182866	1.374488	0.828469	0
13	7	BCL11B	ENSG00000127152	1.351544	0.779631	0
13	8	RTN4	ENSG00000115310	1.337637	0.955331	0
13	9	CD3E	ENSG00000198851	1.32998	0.802263	0
13	10	MYO7B	ENSG00000169994	1.252011	0.695057	0
14	1	CD74	ENSG00000019582	0.660103	0.99457	0
14	2	HLA-DRA	ENSG00000204287	0.735602	0.979186	3.60252e-312
14	3	ENSG00000227706	ENSG00000227706	1.248572	0.826244	4.85246e-252
14	4	ADARB2	ENSG00000185736	1.289892	0.748416	1.47529e-218
14	5	ENSG00000287092	ENSG00000287092	1.028207	0.8181	1.02777e-188
14	6	GNG7	ENSG00000176533	0.867041	0.851584	1.17846e-177
14	7	ENSG00000271758	ENSG00000271758	1.089855	0.708597	1.09347e-176
14	8	PCDH9	ENSG00000184226	1.04408	0.754751	1.37293e-176
14	9	ENSG00000249815	ENSG00000249815	1.06613	0.602715	5.01166e-173
14	10	KCNN1	ENSG00000105642	0.942593	0.650679	2.1271e-170
15	1	ENSG00000260371	ENSG00000260371	2.0081	0.959602	0
15	2	RPS4Y2	ENSG00000280969	1.86301	0.924737	0
15	3	RPL10P9	ENSG00000233913	1.749195	0.913669	0
15	4	TCL1B	ENSG00000213231	1.743447	0.798561	0
15	5	STAG3	ENSG00000066923	1.661438	0.909795	0
15	6	ARPP21	ENSG00000172995	1.472932	0.885169	0
15	7	ENSG00000227706	ENSG00000227706	1.466779	0.882402	0
15	8	ENSG00000259084	ENSG00000259084	1.398711	0.669618	0
15	9	CD79A	ENSG00000105369	1.373332	0.935529	0
15	10	GNG7	ENSG00000176533	1.348525	0.952407	0
16	1	HLA-DRA	ENSG00000204287	1.147663	0.9965	0
16	2	HLA-DPB1	ENSG00000223865	0.998031	0.967629	0
16	3	CD74	ENSG00000019582	0.968972	0.9965	0
16	4	MALAT1	ENSG00000251562	0.451195	1	0
16	5	RPS27	ENSG00000177954	0.416379	0.995626	0
16	6	RPL10	ENSG00000147403	0.338563	0.99825	0
16	7	TPT1	ENSG00000133112	0.286644	0.999125	0
16	8	HLA-DPA1	ENSG00000231389	0.948027	0.951881	3.75115e-312
16	9	RPLP2	ENSG00000177600	0.310874	0.997375	2.03162e-301
16	10	EEF1A1	ENSG00000156508	0.330897	0.9965	3.72462e-291
17	1	HS3ST4	ENSG00000182601	1.709605	0.865854	0
17	2	ENSG00000285534	ENSG00000285534	1.607702	0.963415	0
17	3	MDK	ENSG00000110492	1.210603	0.895379	0
17	4	ENSG00000271758	ENSG00000271758	1.173599	0.799743	0
17	5	NRN1	ENSG00000124785	1.112	0.760591	0
17	6	TCL1A	ENSG00000100721	1.07727	0.955712	0
17	7	RPL9	ENSG00000163682	0.993089	0.996791	0

17	8	GNG7	ENSG00000176533	0.987594	0.954429	0
17	9	CDK14	ENSG00000058091	0.973872	0.912709	0
17	10	CD24	ENSG00000272398	0.96605	0.928113	0
18	1	TUBA1B	ENSG00000123416	1.154398	1	1.40796e-115
18	2	HSP90AA1	ENSG00000080824	1.040893	1	1.0172e-113
18	3	HBA1	ENSG00000206172	2.241851	1	4.84755e-108
18	4	HSPA1A	ENSG00000204389	2.419819	0.990741	4.68026e-98
18	5	DNAJB1	ENSG00000132002	1.943407	0.990741	4.02661e-92
18	6	FTL	ENSG00000087086	0.705953	1	8.27478e-92
18	7	HSPA8	ENSG00000109971	1.03534	1	1.79847e-86
18	8	CA1	ENSG00000133742	2.215093	0.981481	1.09689e-84
18	9	FOS	ENSG00000170345	0.733333	1	3.20514e-81
18	10	AHSP	ENSG00000169877	2.116638	0.981481	5.45986e-81
19	1	JCHAIN	ENSG00000132465	1.489068	0.958026	0
19	2	NRXN3	ENSG00000021645	1.484617	0.982416	0
19	3	CRADD	ENSG00000169372	1.406597	0.972206	0
19	4	ID1	ENSG00000125968	1.36275	0.922859	0
19	5	ID3	ENSG00000117318	1.275162	0.972774	0
19	6	S100A16	ENSG00000188643	1.23019	0.966534	0
19	7	CCDC26	ENSG00000229140	1.047646	0.862167	0
19	8	RPL17-C18orf32	ENSG00000215472	1.041623	0.996029	0
19	9	CD34	ENSG00000174059	1.037241	0.880885	0
19	10	HLA-DRB5	ENSG00000198502	1.011139	0.898469	0
20	1	MTND1P23	ENSG00000225972	1.986608	0.99532	0
20	2	MTCO1P12	ENSG00000237973	1.552166	0.970359	0
20	3	RPL10P9	ENSG00000233913	1.542181	1	0
20	4	WASH6P	ENSG00000182484	1.524067	0.940718	0
20	5	HOPX	ENSG00000171476	1.514714	0.971919	0
20	6	FCER1G	ENSG00000158869	1.500497	0.953198	0
20	7	PRSS57	ENSG00000185198	1.465144	0.99376	0
20	8	ITM2A	ENSG00000078596	1.446974	0.984399	0
20	9	RAB27B	ENSG00000041353	1.404328	0.946958	0
20	10	IGF2BP2	ENSG00000073792	1.319302	0.948518	0
21	1	MTND1P23	ENSG00000225972	1.972847	0.990942	0
21	2	WASH6P	ENSG00000182484	1.583862	0.964674	0
21	3	MTCO1P12	ENSG00000237973	1.539834	0.967391	0
21	4	FCER1G	ENSG00000158869	1.526857	0.961051	0
21	5	LINGO2	ENSG00000174482	1.524661	0.935688	0
21	6	RPL10P9	ENSG00000233913	1.518504	1	0
21	7	HOPX	ENSG00000171476	1.502142	0.977355	0
21	8	PRSS57	ENSG00000185198	1.482948	0.996377	0
21	9	ITM2A	ENSG00000078596	1.481794	0.986413	0
21	10	ITGA2B	ENSG00000005961	1.401856	0.953804	0
22	1	NRXN3	ENSG00000021645	1.506439	0.985678	0
22	2	JCHAIN	ENSG00000132465	1.448449	0.938671	0
22	3	CRADD	ENSG00000169372	1.443066	0.973926	0
22	4	ID1	ENSG00000125968	1.372879	0.918472	0
22	5	ID3	ENSG00000117318	1.299638	0.969152	0
22	6	S100A16	ENSG00000188643	1.262899	0.968417	0
22	7	RPL36A	ENSG00000241343	1.239233	0.995226	0
22	8	CD34	ENSG00000174059	1.151285	0.914433	0

22	9	CCDC26	ENSG00000229140	1.068786	0.869996	0
22	10	HLA-DQA1	ENSG00000196735	1.054722	0.945648	0
23	1	NRXN3	ENSG00000021645	1.52858	0.988428	0
23	2	JCHAIN	ENSG00000132465	1.524302	0.956606	0
23	3	CRADD	ENSG00000169372	1.448371	0.979749	0
23	4	ID1	ENSG00000125968	1.380728	0.922854	0
23	5	ID3	ENSG00000117318	1.281516	0.96866	0
23	6	S100A16	ENSG00000188643	1.253889	0.965767	0
23	7	CD34	ENSG00000174059	1.066007	0.878978	0
23	8	CCDC26	ENSG00000229140	1.059113	0.862584	0
23	9	HLA-DRB5	ENSG00000198502	1.014635	0.895853	0
23	10	HLA-DQA1	ENSG00000196735	1.013505	0.938766	0
24	1	HMGB2	ENSG00000164104	0.994665	1	3.87587e-169
24	2	TUBA1B	ENSG00000123416	0.925013	1	5.34947e-168
24	3	H2AZ1	ENSG00000164032	0.977696	1	4.45797e-156
24	4	HMG2	ENSG00000198830	0.77171	1	2.31669e-143
24	5	STMN1	ENSG00000117632	0.732182	1	3.95909e-133
24	6	S100A16	ENSG00000188643	0.988669	1	5.42265e-124
24	7	RAN	ENSG00000132341	0.720021	1	1.97858e-118
24	8	TUBB	ENSG00000196230	0.752377	1	2.61222e-117
24	9	THSD4-AS1	ENSG00000259964	1.065697	1	6.88758e-116
24	10	PKM	ENSG00000067225	0.774097	1	3.81264e-112

Complete detailed result: [scrna_marker_genes.csv](#) (also indexed in [scRNA Supplementary Results.xlsx](#) and the Full Results Package).

15. Cell-type annotation

Measure	Result
Method	registry_driven_hierarchical_evidence_v1_tnk_competitive_fine_v2
Annotation profile	human_pbmc_v1
Profile version	1.0.0
Profile qualification status	qualification_in_progress
Species	human
Biological context	PBMC / peripheral blood
Gene-symbol resolution used	Yes
Resolved feature IDs	40316
Matrix feature identity preserved	Yes
Assigned cells	25179
Unassigned cells	1127
Unassigned fraction	0.042842
Detected lineages	B lineage, Myeloid lineage, Platelet/Megakaryocyte lineage, T/NK lineage

Boundary	Automated evidence-based cell-type annotation with confidence scoring and expert-review support. This is not definitive ground-truth annotation.
Profile product claim / interpretation policy	Automated evidence-based cell-type annotation with confidence scoring and expert-review support.

Cluster annotations

Cluster	Cell Count	Broad Lineage	Broad Confidence	Fine Annotation	Fine Confidence	Annotation	Assignment Status	Score Margin	Supporting Markers
0	709	T/NK lineage	high	T cell	medium	T cell	resolved	0.096716	TRAC, TRBC1, TRBC2, CD247, CD2, IL7R
1	605	Unassigned	low	—	unresolved	Unassigned	unassigned	0.02741	TUBB1
2	688	B lineage	medium	—	low	B lineage · subtype unresolved	broad_lineage_resolved_subtype_unresolved	0.05829	DERL3
3	414	T/NK lineage	high	T cell	medium	T cell	resolved	-0.152823	CD3D, CD3E, TRAC, TRBC1, TRBC2, CD247, LCK, CD2, CD7, GIMAP7, IL7R, TCF7
4	522	Unassigned	low	—	unresolved	Unassigned	unassigned	0.131652	TUBB1
5	265	Myeloid lineage	medium	Monocyte/macrophage	high	Monocyte/macrophage	resolved	0.132442	LYZ, LST1, FCGR3A, S100A8,

									S100 A9, CTSS , CTS D, LGA LS3, TYR OBP, FCE R1G, CST3 , CTS C, SAT1 , LGA LS9
6	79 0	T/NK lineage	high	T cell	high	T cell	resolved	0.66 7201	CD3 D, CD3 E, TRA C, TRB C1, TRB C2, CD24 7, LCK, CD2, CD7, GIM AP7, LTB, IL7R, CCR 7, TCF7 , LEF1
7	12 72	B lineage	mediu m	Plasma cell	mediu m	Plasma cell	resolved	0.15 0512	MZB 1, XBP1
8	41 5	B lineage	high	B cell	high	B cell	resolved	0.18 3245	MS4 A1, CD79 A, CD79 B, CD22 ,

									CD37 , CD74 , HLA- DRA, HLA- DPA 1, HLA- DPB1 , CD40
9	72	B lineage	high	Plasma cell	high	Plasma cell	resolved	0.16 9689	MZB 1, JCH AIN, DER L3
10	40 6	B lineage	mediu m	Plasma cell	high	Plasma cell	resolved	0.10 4838	MZB 1, JCH AIN, DER L3
11	22 0	B lineage	high	Plasma cell	high	Plasma cell	resolved	0.10 7161	MZB 1, XBP1 , DER L3
12	22 44	T/NK lineage	high	T cell	high	T cell	resolved	0.34 1874	CD3 D, CD3 E, TRA C, TRB C1, TRB C2, CD24 7, LCK, CD7, GIM AP7, LTB, IL7R, TCF7
13	16 79	T/NK lineage	high	T cell	high	T cell	resolved	0.79 6962	CD3 D, CD3 E,

									TRA C, TRB C1, TRB C2, CD24 7, LCK, CD2, CD7, GIM AP7, LTB, IL7R, TCF7 , LEF1
14	11 05	B lineage	high	—	low	B lineage · subtype unresolved	broad_lineage_resolved_ subtype_unresolved	0.10 6484	—
15	36 14	B lineage	high	Plasma cell	medium	Plasma cell	resolved	0.14 8657	JCH AIN, XBP1
16	11 43	B lineage	high	—	low	B lineage · subtype unresolved	broad_lineage_resolved_ subtype_unresolved	0.02 7621	JCH AIN
17	15 58	B lineage	medium	—	low	B lineage · subtype unresolved	broad_lineage_resolved_ subtype_unresolved	0.01 7467	—
18	10 8	T/NK lineage	high	—	unresolved	T/NK lineage · subtype unresolved	broad_lineage_resolved_ subtype_unresolved	0.15 7557	NKG 7, GNL Y, PRF1 , FGF BP2, GZM H
19	17 63	B lineage	high	B cell	medium	B cell	resolved	0.04 0028	MS4 A1, CD19 , HLA- DPA 1, HLA- DPB1 , CD40
20	64 1	Platelet/Meg akaryocyte lineage	high	Platelet	high	Platelet	resolved	0.71 0658	PF4, GP9, ITGA

									2B, RGS1 8, NRG N, GNG 11, SPA RC
21	11 04	Platelet/Meg akaryocyte lineage	high	Platelet	high	Platelet	resolved	0.64 0799	PF4, GP9, ITGA 2B, RGS1 8, NRG N, GNG 11, SPA RC
22	27 23	B lineage	high	B cell	mediu m	B cell	resolved	0.03 5071	MS4 A1, CD19 , HLA- DPA 1, HLA- DPB1 , CD40
23	20 74	B lineage	high	B cell	mediu m	B cell	resolved	0.03 7409	MS4 A1, CD19 , HLA- DPA 1, HLA- DPB1 , CD40
24	17 2	B lineage	mediu m	Plasma cell	high	Plasma cell	resolved	0.16 3542	JCH AIN, XBP1 , DER L3

Complete detailed result: [scrna_cluster_annotations.csv](#) (also indexed in [scRNA Supplementary Results.xlsx](#) and the Full Results Package).

Annotation evidence availability

Leve	Annotation	Availabl	Total	Availabl	Total	Anchor	Evidence	Absen	Absent
------	------------	----------	-------	----------	-------	--------	----------	-------	--------

l		e Anchor Genes	Ancho r Genes	e Support Genes	Support Genes	Availabilit y Fraction	Availabilit y Fraction	t Ancho r Genes From Input	Support Genes From Input
broa d	T/NK lineage	11	11	8	8	1	1	—	—
broa d	B lineage	7	7	6	6	1	1	—	—
broa d	Myeloid lineage	8	8	6	6	1	1	—	—
broa d	Platelet/Megakaryoc yte lineage	6	6	3	4	1	0.9	—	SDPR
fine	T cell	7	7	9	9	1	1	—	—
fine	NK cell	7	7	6	6	1	1	—	—
fine	B cell	6	6	5	5	1	1	—	—
fine	Plasma cell	5	5	3	3	1	1	—	—
fine	Monocyte/macrophage	8	8	6	6	1	1	—	—
fine	Dendritic cell	4	4	6	6	1	1	—	—
fine	Platelet	6	6	3	4	1	0.9	—	SDPR

Missing markers are reported as absent from the current input matrix only. BioFlow does not infer whether they were absent in the original experiment or removed by an upstream/subset-selection step.

Annotation assignment diagnostics

Cl ust er	Ce ll C ou nt	Predicted Broad Lineage	Predicted Annotatio n	Best Broad Candidate	Best Bro ad Sco re	Best Bro ad Suff icie nt	Best Broa d Conf idenc e	Best Bro ad Sco re Mar gin	Best Bro ad Ava ilabl e Anc hors	Bes t Bro ad Tot al An cho rs	Best Bro ad Anc hor Hits	Best Broad Failure Reasons
0	70 9	T/NK lineage	T cell	T/NK lineage	0.23 921 5	Yes	high	0.11 061	11	11	TR AC, TR BC1 , TR BC2 , CD2 47, GN LY	—
1	60 5	Unassigne d	Unassigne d	Platelet/M egakaryoc yte lineage	0.18 162 7	Yes	low	0.02 741	6	6	TU BB1	insufficient_sco re_margin
2	68 8	B lineage	B lineage · subtype unresolve	B lineage	0.13 947 3	Yes	medi um	0.03 941 8	7	7	CD7 9B	—

			d									
3	414	T/NK lineage	T cell	T/NK lineage	0.783233	Yes	high	0.627667	11	11	CD3D, CD3E, TRAC, TRBC1, TRBC2, CD247, NK G7, GNLY, KL RD1, CTSW, PRF1	—
4	522	Unassigned	Unassigned	Platelet/Megakaryocyte lineage	0.220086	Yes	low	0.131652	6	6	TU BB1	insufficient_confidence_evidence
5	265	Myeloid lineage	Monocyte/macrophage	Myeloid lineage	0.646488	Yes	medium	0.040866	8	8	LYZ, LST1, FCGR3A, S100A8, S100A9, CTS S, CTS D, FCE R1G	—
6	790	T/NK lineage	T cell	T/NK lineage	0.589133	Yes	high	0.538141	11	11	CD3D, CD3E, TRAC,	—

											TR BC1 , TR BC2 , CD2 47, CTS W	
7	12 72	B lineage	Plasma cell	B lineage	0.21 743 8	Yes	medi um	0.05 885 2	7	7	MS 4A1 , CD7 9A, CD2 2	—
8	41 5	B lineage	B cell	B lineage	0.45 297 7	Yes	high	0.40 803 6	7	7	MS 4A1 , CD7 9A, CD7 9B, CD2 2, CD3 7, CD7 4	—
9	72	B lineage	Plasma cell	B lineage	0.33 464 2	Yes	high	0.26 169 7	7	7	MS 4A1 , CD7 9A, CD7 9B, CD2 2	—
10	40 6	B lineage	Plasma cell	B lineage	0.20 368	Yes	medi um	0.07 897 5	7	7	MS 4A1 , CD2 2	—
11	22 0	B lineage	Plasma cell	B lineage	0.21 435 2	Yes	high	0.16 321 3	7	7	CD7 9A, CD7 9B, CD3 7, CD7 4	—
12	22 44	T/NK lineage	T cell	T/NK lineage	0.62 787	Yes	high	0.57 135	11	11	CD3 D,	—

					5			1			CD3 E, TR AC, TR BC1 , TR BC2 , CD2 47, NK G7, CTS W	
13	16 79	T/NK lineage	T cell	T/NK lineage	0.46 001 3	Yes	high	0.46 001 3	11	11	CD3 D, CD3 E, TR AC, TR BC1 , TR BC2 , CD2 47	—
14	11 05	B lineage	B lineage · subtype unresolv ed	B lineage	0.17 939 5	Yes	high	0.08 925 5	7	7	CD7 9A, CD2 2, CD7 4	—
15	36 14	B lineage	Plasma cell	B lineage	0.29 502	Yes	high	0.20 193	7	7	MS 4A1 , CD7 9A, CD7 9B, CD1 9, CD7 4	—
16	11 43	B lineage	B lineage · subtype unresolv ed	B lineage	0.19 529 2	Yes	high	0.11 041 1	7	7	MS 4A1 , CD7 9A, CD7 4	—

17	1558	B lineage	B lineage · subtype unresolved	B lineage	0.219539	Yes	medium	0.104872	7	7	CD79A	—
18	108	T/NK lineage	T/NK lineage · subtype unresolved	T/NK lineage	0.235814	Yes	high	0.157557	11	11	CD3D, CD247, NK G7, GN LY, PRF1	—
19	1763	B lineage	B cell	B lineage	0.260872	Yes	high	0.10699	7	7	MS4A1, CD19	—
20	641	Platelet/Megakaryocyte lineage	Platelet	Platelet/Megakaryocyte lineage	0.648877	Yes	high	0.308048	6	6	PF4, GP9, ITGA2B, RGS18	—
21	1104	Platelet/Megakaryocyte lineage	Platelet	Platelet/Megakaryocyte lineage	0.576822	Yes	high	0.23824	6	6	PF4, GP9, ITGA2B, RGS18	—
22	2723	B lineage	B cell	B lineage	0.27677	Yes	high	0.157412	7	7	MS4A1, CD19	—
23	2074	B lineage	B cell	B lineage	0.26759	Yes	high	0.164815	7	7	MS4A1, CD19	—
24	172	B lineage	Plasma cell	B lineage	0.29043	Yes	medium	0.072647	7	7	MS4A1, CD19, CD22	—

Annotation stability across conditions

Annotation	Condition	Cell Count	Biological Samples
B cell	—	105	3
B cell	B_ALL	6823	17
B cell	T_ALL	47	4
B lineage · subtype unresolved	B_ALL	4493	11
B lineage · subtype unresolved	T_ALL	1	1
Monocyte/macrophage	—	138	3
Monocyte/macrophage	B_ALL	68	15
Monocyte/macrophage	T_ALL	59	4
Plasma cell	B_ALL	5754	16
Plasma cell	T_ALL	2	1
Platelet	—	1745	1
T cell	—	331	3
T cell	B_ALL	1432	17
T cell	T_ALL	4073	4
T/NK lineage · subtype unresolved	B_ALL	108	2
Unassigned	—	871	3
Unassigned	B_ALL	242	18
Unassigned	T_ALL	14	4

Annotation quality gate for downstream DA

Measure					Result			
Profile qualification status					qualification in progress			
Any annotation-dependent inference publishable					No			
Gate summary					annotation profile not qualified			
Annotation	Status	Reason	Qualification Level	Qualification Status	Conditions Present	Conditions Expected	Low Confidence Cell Fraction	Publishable
B cell	blocked	annotation_profile_not_qualified	profile	NOT_QUALIFIED	2	2	0	No
B lineage · subtype unresolved	blocked	annotation_profile_not_qualified	profile	NOT_QUALIFIED	2	2	0	No
Monocyte/macrophage	blocked	annotation_profile_not_qualified	profile	NOT_QUALIFIED	2	2	0	No
Plasma cell	blocked	annotation_profile_not_qualified	profile	NOT_QUALIFIED	2	2	0	No
Platelet	blocked	annotation_profile_not_qualified	profile	NOT_QUALIFIED	0	2	0	No
T cell	blocked	annotation_profile_not_qualified	profile	NOT_QUALIFIED	2	2	0	No
T/NK lineage · subtype unresolved	blocked	annotation_profile_not_qualified	profile	NOT_QUALIFIED	1	2	0	No
Unassigned	blocked	unassigned_population	none	BLOCKED	0	2	—	No

CellTypist T/NK fine annotation candidate

Measure	Result
Adapter	scrna.annotation.fine.celltypist_tnk
Adapter version	1.0.0
Candidate status	Optional fine-annotation runtime not active; main workflow unaffected
Active for customer results	No
Model	Immune All Low.pkl

CellTypist candidate was not run because the BioFlow-managed runtime/model is not ready. Existing BioFlow annotations are unchanged.

Annotation stability notes

- Prediction is registry-driven and tissue/species-aware. Condition labels and hidden reference labels do not participate in annotation scoring.
- Broad lineage is resolved before fine subtype; when subtype evidence is insufficient BioFlow preserves an explicit unresolved broad-lineage state rather than forcing a fine label.
- This post-annotation QC gate never changes predicted labels. Populations observed in only one condition or dominated by low-confidence assignments are blocked from publishable differential-abundance interpretation.

16. Cell composition

Measure	Result
Available	Yes
Method	biological_sample_cell_population_composition_v1
Raw aggregation unit	sample_id
Biological samples	25
Cell populations	8
Conditions	B ALL, T ALL

Condition-level composition summary

Condition	Annotation	Biological Samples	Mean Percent	Median Proportion	Sd Proportion
B ALL	B cell	18	14.024519	0.015699	0.304477
B_ALL	B lineage · subtype unresolved	18	30.372702	0.021021	0.382582
B ALL	Monocyte/macrophage	18	2.761479	0.002729	0.101716
B ALL	Plasma cell	18	39.130314	0.025171	0.443584
B ALL	Platelet	18	0	0	0
B ALL	T cell	18	8.093341	0.051833	0.114848
B_ALL	T/NK lineage · subtype unresolved	18	1.532472	0	0.064823
B ALL	Unassigned	18	4.085175	0.010767	0.072353
T ALL	B cell	4	1.745769	0.011322	0.018873
T_ALL	B lineage · subtype unresolved	4	0.280899	0	0.005618
T ALL	Monocyte/macrophage	4	1.903868	0.014484	0.018389
T ALL	Plasma cell	4	0.112108	0	0.002242
T ALL	Platelet	4	0	0	0
T ALL	T cell	4	94.350271	0.972384	0.067721
T_ALL	T/NK lineage · subtype unresolved	4	0	0	0

T_ALL	Unassigned	4	1.607085	0.003753	0.026787
-------	------------	---	----------	----------	----------

Complete detailed result: scrna_cell_composition_condition_summary.csv (also indexed in scRNA Supplementary Results.xlsx and the Full Results Package).

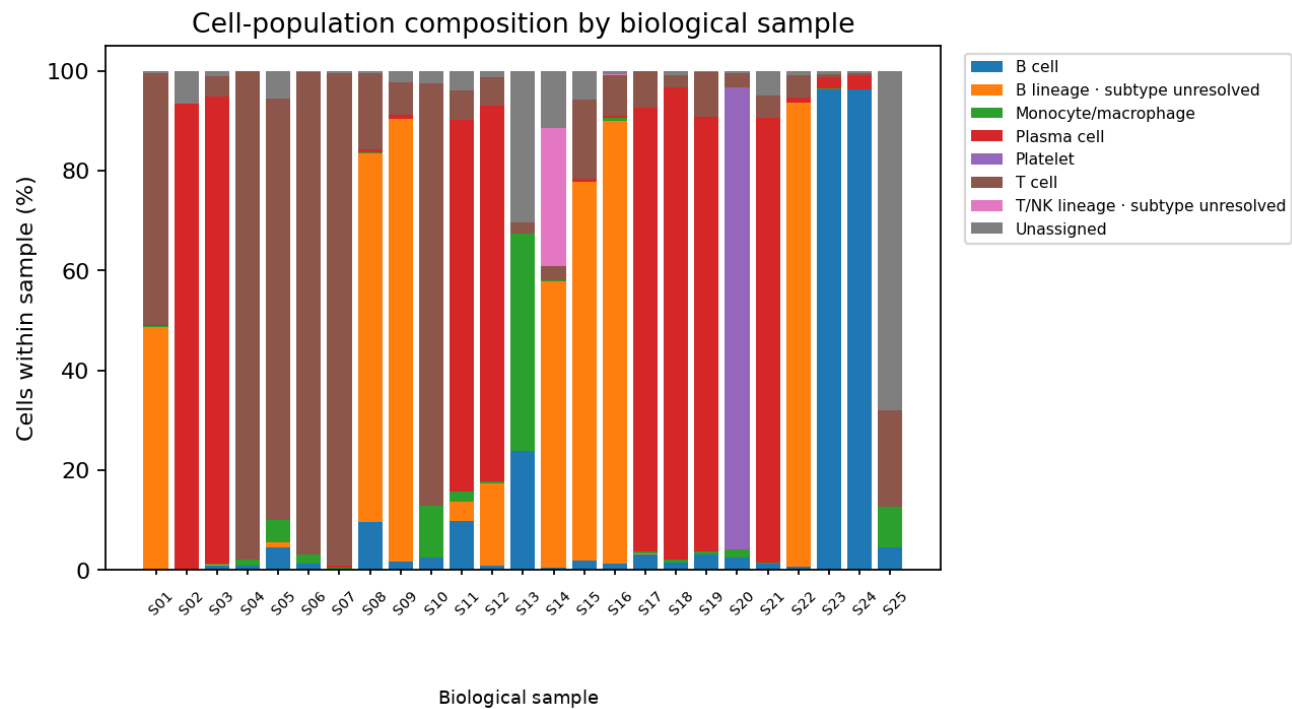


Figure 6. Cell-population composition within each biological sample. Condition-level inference does not pool cells across samples.

Cell-composition notes

- Cell-type proportions are calculated within each biological sample; pooled condition-level cell counts are not used as independent replicates.
- A zero count for a sample × cell population is retained as a real composition observation rather than silently dropping the sample.

17. Differential abundance

Measure	Result
Available	No
Method	CLR linear model after experimental-unit resolution
Raw aggregation unit	sample_id
Effective independent replicate unit	subject_id
Blocking unit	none
Inference status	INSUFFICIENT_INDEPENDENT_REPLICATES
Feasibility reason code	INSUFFICIENT_INDEPENDENT_REPLICATES
Feasibility reason	Inference is blocked because at least one contrast group has fewer than the required number of independent experimental units.
Contrast	T_ALL vs B_ALL
Raw units by group	{'T_ALL': 4, 'B_ALL': 18}
Independent units by group	{'T_ALL': 2, 'B_ALL': 15}
Minimum independent units required per group	3

Cell populations tested	0
Blocked by annotation QC	0
Annotation technical gate	PARTIAL
Annotation qualification gate	NOT_QUALIFIED
Annotation profile qualification status	qualification_in_progress
Publishable annotation-dependent inference	No
Annotation gate reason	annotation_profile_not_qualified
Publishable DA inference	No
Significant FDR < 0.05	0
Batch adjustment	Not required
Transform	centered log-ratio of inference-unit cell-population proportions
Multiple testing	Benjamini-Hochberg FDR across cell populations

Differential-abundance results

No rows were produced.

Complete result artifact: `scrna_differential_abundance.csv`.

Differential-abundance interpretation notes

- Independent-replicate resolution status: REPEATED_SUBJECT_NESTED_WITHIN_CONDITION; effective replicate unit: `subject_id`.
- Cell-composition inference and pseudobulk DE use the same contrast-specific experimental-unit contract; individual cells are never condition replicates.
- The model tests centered-log-ratio (CLR) transformed inference-unit compositions.
- A 0.5-cell zero pseudocount is used only for the CLR transform; reported percentages use observed counts.
- BH-FDR is applied across tested cell populations for the selected condition contrast.
- Repeated biological samples are nested within one condition per subject; safe inference uses subject-level collapse.
- Inference is blocked because at least one contrast group has fewer than the required number of independent experimental units.

18. Pseudobulk aggregation

Measure	Result
Available	Yes
Method	sum raw counts by biological sample and cell annotation v1
Raw aggregation unit	sample_id
Groups	127
Samples	25
Cell populations	8
Minimum cells per pseudobulk	10

Cell-population pseudobulk summary

Annotation	Biological Samples	Pseudobulk Groups	Usable Groups	Total Cells	Median Cells Per Group
B cell	24	24	13	6975	10.5
B lineage · subtype unresolved	12	12	8	4494	115
Monocyte/macrophage	22	22	5	265	4
Plasma cell	17	17	11	5756	67

Platelet	1	1	1	1745	1745
T cell	24	24	21	5836	42.5
T/NK lineage · subtype unresolved	2	2	1	108	54
Unassigned	25	25	9	1127	7

Complete detailed result: `scrna_pseudobulk_group_summary.csv` (also indexed in `scRNA Supplementary Results.xlsx` and the Full Results Package).

Population-level replicate capacity before condition contrast

Annotation	Usable Groups	Minimum Replicates Per Group	Minimum Total Groups For Two Groups	Precondition Status
B cell	13	3	6	CONDITION_SPLIT_REQUIRES_FINAL_READINESS_CHECK
B lineage · subtype unresolved	8	3	6	CONDITION_SPLIT_REQUIRES_FINAL_READINESS_CHECK
Monocyte/macrophage	5	3	6	INSUFFICIENT_TOTAL_CAPACITY
Plasma cell	11	3	6	CONDITION_SPLIT_REQUIRES_FINAL_READINESS_CHECK
Platelet	1	3	6	INSUFFICIENT_TOTAL_CAPACITY
T cell	21	3	6	CONDITION_SPLIT_REQUIRES_FINAL_READINESS_CHECK
T/NK lineage · subtype unresolved	1	3	6	INSUFFICIENT_TOTAL_CAPACITY
Unassigned	9	3	6	EXPLORATORY_UNASSIGNED

This is a necessary-capacity audit only. A population with fewer than twice the minimum replicate count can never support two condition groups at the configured minimum. Populations that pass this preview are not DE-ready until the actual condition split is supplied and the existing per-condition readiness gate passes.

Persisted generic capacity audit: total usable sample×population groups are a necessary condition only; actual condition-specific readiness is decided by the experimental-unit and independent-replicate gate.

Pseudobulk notes

- Raw counts are summed within biological sample × annotation groups before any condition-level comparison.
- Small sample×annotation groups are retained in the artifact but excluded from inferential screening when below the configured cell-count floor.

19. Pseudobulk differential expression

Measure	Result
Available	No
Method	edger
Engine	edgeR quasi-likelihood GLM
Raw aggregation unit	sample_id
Effective independent replicate unit	subject_id
Blocking unit	none
Inference status	INSUFFICIENT_INDEPENDENT_REPLICATES
Feasibility reason code	INSUFFICIENT_INDEPENDENT_REPLICATES
Feasibility reason	Inference is blocked because at least one contrast group has

	fewer than the required number of independent experimental units.
Design structure status	REPEATED SUBJECT NESTED WITHIN CONDITION
Raw units by group	{'T_ALL': 4, 'B_ALL': 18}
Independent units by group	{'T_ALL': 2, 'B_ALL': 15}
Minimum independent units required per group	3
Repeated subjects	5
Subjects nested within one condition	5
Annotation technical gate	PARTIAL
Annotation qualification gate	NOT_QUALIFIED
Annotation profile qualification status	qualification in progress
Publishable annotation-dependent inference	No
Annotation gate reason	annotation profile not qualified
Publishable DE inference	No
Contrast	T_ALL vs B_ALL
Positive direction	Positive log2 FC = higher in T_ALL
Batch adjustment	Not required
Design columns	—
Mechanical tests retained	0
Mechanical significant FDR < 0.05	0
Publishable tests	0
Publishable significant FDR < 0.05	0
Rows truncated upstream	0

Pseudobulk DE readiness

Annotation	Raw Units A	Independent Units A	Raw Units B	Independent Units B	Effective Replicate Unit	Minimum Required	Design Is Estimable	Ready	Readiness Reason Code	Readiness Message
B cell	2	2	9	7	subject_id	3	—	No	INSUFFICIENT_POPULATION_CAPACITY	This cell population has too few usable independent experimental units for the configured replicate requirement.
B lineage · subtype unresolved	0	0	8	7	subject_id	3	—	No	INSUFFICIENT_POPULATION_CAPACITY	This cell population has too few

										usable independent experimental units for the configured replicate requirement.
Monocyte/macrophage	2	2	1	2	subject_id	3	—	No	INSUFFICIENT_POPULATION_CAPACITY	This cell population has too few usable independent experimental units for the configured replicate requirement.
Plasma cell	0	0	11	9	subject_id	3	—	No	INSUFFICIENT_POPULATION_CAPACITY	This cell population has too few usable independent experimental units for the configured replicate requirement.
T cell	4	2	14	12	subject_id	3	—	No	INSUFFICIENT_POPULATION_CAPACITY	This cell population has too few usable

										independent experimental units for the configured replicate requirement.
T/NK lineage · subtype unresolved	0	0	1	1	subject_id	3	—	No	INSUFFICIENT_POPULATION_CAPACITY	This cell population has too few usable independent experimental units for the configured replicate requirement.
Unassigned	0	0	8	7	subject_id	3	—	No	INSUFFICIENT_POPULATION_CAPACITY	This cell population has too few usable independent experimental units for the configured replicate requirement.

Complete detailed result: scrna_pseudobulk_de_readiness.csv (also indexed in scRNA Supplementary Results.xlsx and the Full Results Package).

DE summary by cell population

No rows were produced.

Top 20 DE genes per cell population

No rows were produced.

Complete result artifact: `scrna_pseudobulk_condition_de.csv`.

Pseudobulk DE interpretation notes

- Sample-level raw-count pseudobulks are retained unchanged; inferential replication is resolved separately at subject_id level.
- Independent-replicate resolution status: REPEATED_SUBJECT_NESTED_WITHIN_CONDITION; raw sample counts and effective independent-unit counts are reported separately.
- edgeR quasi-likelihood GLM uses raw counts and a contrast-specific design validated before R execution.
- Repeated biological samples are nested within one condition per subject; safe inference uses subject-level collapse.
- Count-model fits are retained for audit, but annotation-dependent cell-type-specific pseudobulk DE is not publishable because the annotation profile/benchmark qualification gate is not satisfied.
- Unassigned pseudobulk groups may be retained for exploratory audit, but are never publishable cell-type-specific differential-expression populations.
- Inference is blocked because at least one contrast group has fewer than the required number of independent experimental units.

20. Reactome pathway enrichment

Measure	Result
Method	offline reactome hypergeometric ora bh v1
Resource	Reactome
Version	2026-08-11
SHA-256	Omitted in portfolio version
Eligible cluster-level pathway tests	1203
Aggregate hits passing current within-cluster FDR rule	995
Aggregate hit fraction	0.827099
Returned top-hit rows	602
Unique pathways in returned rows	195
Returned Unassigned rows	50
Returned rows with ribosomal-dominant overlap	159
Multiple-testing scope	BH FDR is applied within each cluster over pathways meeting the configured minimum-overlap gate
Aggregate-count scope	Counts are summed across cluster-level tests; they are not a single global pathway FDR family

Reactome method audit

Measure	Result
Input marker method	de evidence ranked markers v1
Positive/selective marker input	Yes
Marker thresholds	{'max_fdr': 0.05, 'min_log_expression_delta': 0.25, 'min_pct_expressing': 0.1}
Exploratory fallback clusters	—
Gene universe definition	resolved gene symbols for all features in the Step-18 log-expression matrix, intersected with each Reactome gene set
Multiple-testing family	within cluster among min overlap eligible pathways

Scientific method qualification	pending
Scientific engine modified by audit	No

Returned top ORA hits by cell-population label

Cell Population	Returned Top Hits	Returned Hits Fdr Lt 0 05
B cell	100	97
B lineage · subtype unresolved	87	68
Monocyte/macrophage	25	25
Plasma cell	144	124
Platelet	46	39
T cell	125	125
T/NK lineage · subtype unresolved	25	25
Unassigned	50	50

Per-population rows below summarize returned top hits, not all pathways tested for that population

Top 10 pathways per cell population

Annotati on	Cl ust er	Pathway	Ov erl ap	P Val ue	Fdr	Overlap Genes
B cell	8	Peptide chain elongation	17	1.2 281 1e- 38	3.2 293 1e- 37	FAU;RPL13;RPL13A;RPL19;RPL23A;RPL30;RPL32;RPL34;RPL36;RPLP1;RPLP2;RPS16;RPS21;RPS25;RPS27;RPS29;RPS6
B cell	8	Viral mRNA Translation	17	1.2 281 1e- 38	3.2 293 1e- 37	FAU;RPL13;RPL13A;RPL19;RPL23A;RPL30;RPL32;RPL34;RPL36;RPLP1;RPLP2;RPS16;RPS21;RPS25;RPS27;RPS29;RPS6
B cell	8	PELO:HB S1L and ABCE1 dissociate a ribosome on a non-stop mRNA	17	1.5 137 4e- 38	3.2 293 1e- 37	FAU;RPL13;RPL13A;RPL19;RPL23A;RPL30;RPL32;RPL34;RPL36;RPLP1;RPLP2;RPS16;RPS21;RPS25;RPS27;RPS29;RPS6
B cell	8	Eukaryotic Translation Termination	17	2.2 823 1e- 38	3.5 739 5e- 37	FAU;RPL13;RPL13A;RPL19;RPL23A;RPL30;RPL32;RPL34;RPL36;RPLP1;RPLP2;RPS16;RPS21;RPS25;RPS27;RPS29;RPS6
B cell	8	Selenocysteine synthesis	17	2.7 921 5e- 38	3.5 739 5e- 37	FAU;RPL13;RPL13A;RPL19;RPL23A;RPL30;RPL32;RPL34;RPL36;RPLP1;RPLP2;RPS16;RPS21;RPS25;RPS27;RPS29;RPS6
B cell	8	Eukaryotic Translation Elongation	17	3.4 077 7e- 38	3.6 349 5e- 37	FAU;RPL13;RPL13A;RPL19;RPL23A;RPL30;RPL32;RPL34;RPL36;RPLP1;RPLP2;RPS16;RPS21;RPS25;RPS27;RPS29;RPS6
B cell	8	Nonsense	17	4.1	3.7	FAU;RPL13;RPL13A;RPL19;RPL23A;RPL30;RPL32;RPL34;RPL36;RPLP1;RPLP2;RPS16;RPS21;RPS25;RPS27;RPS29;RPS6

		Mediated Decay (NMD) independent of the Exon Junction Complex (EJC)		494 7e-38	938 e-37	L36;RPLP1;RPLP2;RPS16;RPS21;RPS25;RPS27;RPS29;RPS6
B cell	8	ZNF598 and the Ribosome - associated Quality Trigger (RQT) complex dissociate a ribosome stalled on a no-go mRNA	17	5.0 411 7e-38	4.0 329 3e-37	FAU;RPL13;RPL13A;RPL19;RPL23A;RPL30;RPL32;RPL34;RPL36;RPLP1;RPLP2;RPS16;RPS21;RPS25;RPS27;RPS29;RPS6
B cell	8	Formation of a pool of free 40S subunits	17	1.2 918 4e-37	8.2 677 9e-37	FAU;RPL13;RPL13A;RPL19;RPL23A;RPL30;RPL32;RPL34;RPL36;RPLP1;RPLP2;RPS16;RPS21;RPS25;RPS27;RPS29;RPS6
B cell	8	Response of EIF2AK4 (GCN2) to amino acid deficiency	17	1.2 918 4e-37	8.2 677 9e-37	FAU;RPL13;RPL13A;RPL19;RPL23A;RPL30;RPL32;RPL34;RPL36;RPLP1;RPLP2;RPS16;RPS21;RPS25;RPS27;RPS29;RPS6
B lineage · subtype unresolved	16	Peptide chain elongation	9	1.6 847 4e-17	1.0 407 2e-15	EEF1A1;RPL10;RPL28;RPL37;RPL41;RPLP2;RPS15A;RPS18;RPS27
B lineage · subtype unresolved	16	Eukaryotic Translation Elongation	9	2.8 127 6e-17	1.0 407 2e-15	EEF1A1;RPL10;RPL28;RPL37;RPL41;RPLP2;RPS15A;RPS18;RPS27
B lineage · subtype unresolved	16	Viral mRNA Translation	8	3.4 124 4e-15	5.9 062 5e-14	RPL10;RPL28;RPL37;RPL41;RPLP2;RPS15A;RPS18;RPS27
B lineage · subtype unresolved	16	PELO:HB S1L and ABCE1 dissociate	8	3.7 435 4e-15	5.9 062 5e-14	RPL10;RPL28;RPL37;RPL41;RPLP2;RPS15A;RPS18;RPS27

		a ribosome on a non- stop mRNA				
B lineage · subtype unresolv ed	16	Eukaryoti c Translatio n Terminati on	8	4.4 908 7e- 15	5.9 062 5e- 14	RPL10;RPL28;RPL37;RPL41;RPLP2;RPS15A;RPS18;RPS27
B lineage · subtype unresolv ed	16	Selenocyst eine synthesis	8	4.9 111 5e- 15	5.9 062 5e- 14	RPL10;RPL28;RPL37;RPL41;RPLP2;RPS15A;RPS18;RPS27
B lineage · subtype unresolv ed	16	Nonsense Mediated Decay (NMD) independe nt of the Exon Junction Complex (EJC)	8	5.8 558 9e- 15	5.9 062 5e- 14	RPL10;RPL28;RPL37;RPL41;RPLP2;RPS15A;RPS18;RPS27
B lineage · subtype unresolv ed	16	ZNF598 and the Ribosome - associated Quality Trigger (RQT) complex dissociate a ribosome stalled on a no-go mRNA	8	6.3 851 3e- 15	5.9 062 5e- 14	RPL10;RPL28;RPL37;RPL41;RPLP2;RPS15A;RPS18;RPS27
B lineage · subtype unresolv ed	16	Formation of a pool of free 40S subunits	8	9.7 078 e- 15	7.1 837 7e- 14	RPL10;RPL28;RPL37;RPL41;RPLP2;RPS15A;RPS18;RPS27
B lineage · subtype unresolv ed	16	Response of EIF2AK4 (GCN2) to amino acid deficiency	8	9.7 078 e- 15	7.1 837 7e- 14	RPL10;RPL28;RPL37;RPL41;RPLP2;RPS15A;RPS18;RPS27
Monocyt e/macrop	5	Neutrophil degranulat	11	2.5 800	1.1 352	CD44;CD63;CFD;CST3;FCER1G;FTL;LYZ;NPC2;S100A11;S100A9;TYROBP

hage		ion		1e-14	e-12	
Monocyte/macrophage	5	Innate Immune System	12	5.09706e-12	1.12135e-10	CD44;CD63;CFD;CST3;FCER1G;FTL;LYN;LYZ;NPC2;S100A11;S100A9;TYROBP
Monocyte/macrophage	5	Immune System	15	1.09694e-11	1.60885e-10	ANXA1;CD44;CD63;CFD;CST3;FCER1G;FTL;IFI30;LYN;LYZ;NPC2;S100A11;S100A9;TYROBP;VIM
Monocyte/macrophage	5	Hemostasis	7	3.96979e-07	4.36677e-06	CD44;CD63;CFD;FCER1G;LYN;S100A10;SRGN
Monocyte/macrophage	5	Platelet activation, signaling and aggregation	5	1.60037e-06	1.40833e-05	CD63;CFD;FCER1G;LYN;SRGN
Monocyte/macrophage	5	Platelet Adhesion to exposed collagen	2	7.40736e-05	0.000543	FCER1G;LYN
Monocyte/macrophage	5	Platelet degranulation	3	0.000124	0.000686	CD63;CFD;SRGN
Monocyte/macrophage	5	Response to elevated platelet cytosolic Ca ²⁺	3	0.00014	0.000686	CD63;CFD;SRGN
Monocyte/macrophage	5	Dectin-2 family	2	0.000129	0.000686	FCER1G;LYN
Monocyte/macrophage	5	Cell surface interactions at the vascular wall	3	0.000254	0.001115	CD44;FCER1G;LYN
Plasma cell	9	Interferon Signaling	9	2.30269e-12	3.21829e-11	HLA-DPA1;HLA-DPB1;HLA-DRA;HLA-DRB1;HLA-DRB5;IFITM1;IFITM2;IFITM3;RPS4Y2
Plasma cell	9	Translocation of ZAP-70 to Immunological synapse	5	3.38768e-12	3.21829e-11	HLA-DPA1;HLA-DPB1;HLA-DRA;HLA-DRB1;HLA-DRB5
Plasma cell	9	Phosphorylation of CD3 and	5	7.34059e-	4.64904e-	HLA-DPA1;HLA-DPB1;HLA-DRA;HLA-DRB1;HLA-DRB5

		TCR zeta chains		12	11	
Plasma cell	9	Generation of second messenger molecules	5	7.03644e-11	3.34231e-10	HLA-DPA1;HLA-DPB1;HLA-DRA;HLA-DRB1;HLA-DRB5
Plasma cell	9	MHC class II antigen presentation	6	4.87915e-10	1.85408e-09	CD74;HLA-DPA1;HLA-DPB1;HLA-DRA;HLA-DRB1;HLA-DRB5
Plasma cell	9	Cytokine Signaling in Immune system	9	6.64246e-09	1.87867e-08	HLA-DPA1;HLA-DPB1;HLA-DRA;HLA-DRB1;HLA-DRB5;IFITM1;IFITM2;IFITM3;RPS4Y2
Plasma cell	9	Downstream TCR signaling	5	6.92141e-09	1.87867e-08	HLA-DPA1;HLA-DPB1;HLA-DRA;HLA-DRB1;HLA-DRB5
Plasma cell	9	Processing of antigen in germinal center B cells	5	8.73967e-09	2.07567e-08	HLA-DPA1;HLA-DPB1;HLA-DRA;HLA-DRB1;HLA-DRB5
Plasma cell	9	Interferon gamma signaling	5	1.21605e-08	2.56721e-08	HLA-DPA1;HLA-DPB1;HLA-DRA;HLA-DRB1;HLA-DRB5
Plasma cell	9	TCR signaling	5	2.21685e-08	4.21201e-08	HLA-DPA1;HLA-DPB1;HLA-DRA;HLA-DRB1;HLA-DRB5
Platelet	20	Neutrophil degranulation	5	2.95633e-05	0.00031	BST2;CD44;CD63;FCER1G;IMPDH2
Platelet	20	Insulin-like Growth Factor-2 mRNA Binding Proteins (IGF2BPs/IMPs/VICKZs) bind RNA	2	1.73522e-05	0.00031	CD44;IGF2BP2
Platelet	20	Platelet activation, signaling and aggregation	4	4.74701e-05	0.00032	CD63;FCER1G;RAB27B;TAGLN2

		n				
Platelet	20	Hemostasis	5	0.00013	0.000491	CD44;CD63;FCER1G;RAB27B;TAGLN2
Platelet	20	Platelet degranulation	3	0.000124	0.000491	CD63;RAB27B;TAGLN2
Platelet	20	Response to elevated platelet cytosolic Ca ²⁺	3	0.00014	0.000491	CD63;RAB27B;TAGLN2
Platelet	21	Insulin-like Growth Factor-2 mRNA Binding Proteins (IGF2BPs/IMPs/VICKZs) bind RNA	2	1.73522e-05	0.000521	CD44;IGF2BP2
Platelet	21	Hemostasis	5	0.00013	0.000601	CD44;CTSW;FCER1G;ITGA2B;RAB27B
Platelet	21	Platelet activation, signaling and aggregation	4	4.74701e-05	0.000601	CTSW;FCER1G;ITGA2B;RAB27B
Platelet	21	Extracellular matrix organization	4	9.42168e-05	0.000601	CD44;ITGA2B;NCAM1;PRSS2
T cell	6	Peptide chain elongation	21	3.84999e-51	2.31e-49	EEF1A1;RPL11;RPL12;RPL13;RPL28;RPL29;RPL30;RPL32;RPL34;RPL37;RPL37A;RPL41;RPLP1;RPLP2;RPS12;RPS16;RPS18;RPS25;RPS27;RPS29;RPS6
T cell	6	Eukaryotic Translation Elongation	21	1.40609e-50	4.21828e-49	EEF1A1;RPL11;RPL12;RPL13;RPL28;RPL29;RPL30;RPL32;RPL34;RPL37;RPL37A;RPL41;RPLP1;RPLP2;RPS12;RPS16;RPS18;RPS25;RPS27;RPS29;RPS6
T cell	6	Viral mRNA Translation	20	7.26033e-48	1.39999e-46	RPL11;RPL12;RPL13;RPL28;RPL29;RPL30;RPL32;RPL34;RPL37;RPL37A;RPL41;RPLP1;RPLP2;RPS12;RPS16;RPS18;RPS25;RPS27;RPS29;RPS6
T cell	6	PELO:HB S1L and ABCE1 dissociate	20	9.33328e-48	1.39999e-46	RPL11;RPL12;RPL13;RPL28;RPL29;RPL30;RPL32;RPL34;RPL37;RPL37A;RPL41;RPLP1;RPLP2;RPS12;RPS16;RPS18;RPS25;RPS27;RPS29;RPS6

		a ribosome on a non- stop mRNA				
T cell	6	Eukaryoti c Translatio n Terminati on	20	1.5 280 6e- 47	1.8 336 7e- 46	RPL11;RPL12;RPL13;RPL28;RPL29;RPL30;RPL32;RPL34;RPL37;RPL37A;RPL41;RPLP1;RPLP2;RPS12;RPS16;RPS18;RPS25;RPS27;RPS29;RPS6
T cell	6	Selenocyst eine synthesis	20	1.9 464 e- 47	1.9 464 e- 46	RPL11;RPL12;RPL13;RPL28;RPL29;RPL30;RPL32;RPL34;RPL37;RPL37A;RPL41;RPLP1;RPLP2;RPS12;RPS16;RPS18;RPS25;RPS27;RPS29;RPS6
T cell	6	Nonsense Mediated Decay (NMD) independe nt of the Exon Junction Complex (EJC)	20	3.1 308 2e- 47	2.6 835 6e- 46	RPL11;RPL12;RPL13;RPL28;RPL29;RPL30;RPL32;RPL34;RPL37;RPL37A;RPL41;RPLP1;RPLP2;RPS12;RPS16;RPS18;RPS25;RPS27;RPS29;RPS6
T cell	6	ZNF598 and the Ribosome - associated Quality Trigger (RQT) complex dissociate a ribosome stalled on a no-go mRNA	20	3.9 541 1e- 47	2.9 655 8e- 46	RPL11;RPL12;RPL13;RPL28;RPL29;RPL30;RPL32;RPL34;RPL37;RPL37A;RPL41;RPLP1;RPLP2;RPS12;RPS16;RPS18;RPS25;RPS27;RPS29;RPS6
T cell	6	Formation of a pool of free 40S subunits	20	1.2 214 7e- 46	7.3 288 e- 46	RPL11;RPL12;RPL13;RPL28;RPL29;RPL30;RPL32;RPL34;RPL37;RPL37A;RPL41;RPLP1;RPLP2;RPS12;RPS16;RPS18;RPS25;RPS27;RPS29;RPS6
T cell	6	Response of EIF2AK4 (GCN2) to amino acid deficiency	20	1.2 214 7e- 46	7.3 288 e- 46	RPL11;RPL12;RPL13;RPL28;RPL29;RPL30;RPL32;RPL34;RPL37;RPL37A;RPL41;RPLP1;RPLP2;RPS12;RPS16;RPS18;RPS25;RPS27;RPS29;RPS6
T/NK lineage	18	Cellular responses	13	1.9 587	2.4 288	BLVRB;CACNB2;DNAJB1;FOS;HBA1;HSP90AA1;HSPA1A;HSPA8;MT-CO1;RPL8;RPS4X;RPS5;TUBA1B

subtype unresolved		to stimuli		5e-14	5e-12	
T/NK lineage · subtype unresolved	18	Cellular responses to stress	12	1.5138e-13	9.38557e-12	BLVRB;DNAJB1;FOS;HBA1;HSP90AA1;HSPA1A;HSPA8;MT-CO1;RPL8;RPS4X;RPS5;TUBA1B
T/NK lineage · subtype unresolved	18	HSP90 chaperone cycle for steroid hormone receptors (SHR) in the presence of ligand	5	6.7852e-10	2.80455e-08	DNAJB1;HSP90AA1;HSPA1A;HSPA8;TUBA1B
T/NK lineage · subtype unresolved	18	Attenuation phase	4	5.64799e-09	1.75088e-07	DNAJB1;HSP90AA1;HSPA1A;HSPA8
T/NK lineage · subtype unresolved	18	HSF1-dependent transactivation	4	2.11401e-08	5.24275e-07	DNAJB1;HSP90AA1;HSPA1A;HSPA8
T/NK lineage · subtype unresolved	18	Cytokine Signaling in Immune system	8	1.35565e-07	2.42073e-06	CA1;FOS;HSP90AA1;HSPA1A;HSPA8;RPS4X;RPS5;TUBA1B
T/NK lineage · subtype unresolved	18	Axon guidance	7	1.50716e-07	2.42073e-06	CACNB2;HSP90AA1;HSPA8;RPL8;RPS4X;RPS5;TUBA1B
T/NK lineage · subtype unresolved	18	Influenza Infection	5	1.56176e-07	2.42073e-06	HSP90AA1;HSPA1A;RPL8;RPS4X;RPS5
T/NK lineage · subtype unresolved	18	Nervous system development	7	2.057e-07	2.83409e-06	CACNB2;HSP90AA1;HSPA8;RPL8;RPS4X;RPS5;TUBA1B
T/NK lineage · subtype unresolved	18	Antimicrobial mechanism of IFN-stimulated genes	5	7.69642e-07	9.54356e-06	HSPA1A;HSPA8;RPS4X;RPS5;TUBA1B

Unassigned	1	Erythrocytes take up oxygen and release carbon dioxide	6	3.96658e-18	1.22964e-16	CA1;CA2;HBA1;HBB;RHAG;SLC4A1
Unassigned	1	Erythrocytes take up carbon dioxide and release oxygen	6	1.30623e-16	1.34977e-15	CA1;CA2;HBA1;HBB;RHAG;SLC4A1
Unassigned	1	O2/CO2 exchange in erythrocytes	6	1.30623e-16	1.34977e-15	CA1;CA2;HBA1;HBB;RHAG;SLC4A1
Unassigned	4	Erythrocytes take up oxygen and release carbon dioxide	4	2.27604e-11	8.42133e-10	CA1;HBA1;HBB;SLC4A1
Unassigned	4	Erythrocytes take up carbon dioxide and release oxygen	4	1.606e-10	1.98074e-09	CA1;HBA1;HBB;SLC4A1
Unassigned	4	O2/CO2 exchange in erythrocytes	4	1.606e-10	1.98074e-09	CA1;HBA1;HBB;SLC4A1
Unassigned	1	Metabolism of porphyrins	4	3.42525e-09	2.65457e-08	ALAS2;BLVRB;HMBS;UROD
Unassigned	1	Heme biosynthesis	3	1.66518e-07	1.03241e-06	ALAS2;HMBS;UROD
Unassigned	1	Heme assimilation	2	6.2156e-07	3.21139e-06	HBA1;HBB
Unassigned	1	Transport of small molecules	7	7.72326e-07	3.4203e-06	BSG;CA1;CA2;HBA1;HBB;RHAG;SLC4A1

Complete detailed result: `scrna_reactome_enrichment.csv` (also indexed in `scRNA Supplementary Results.xlsx` and the Full Results Package).

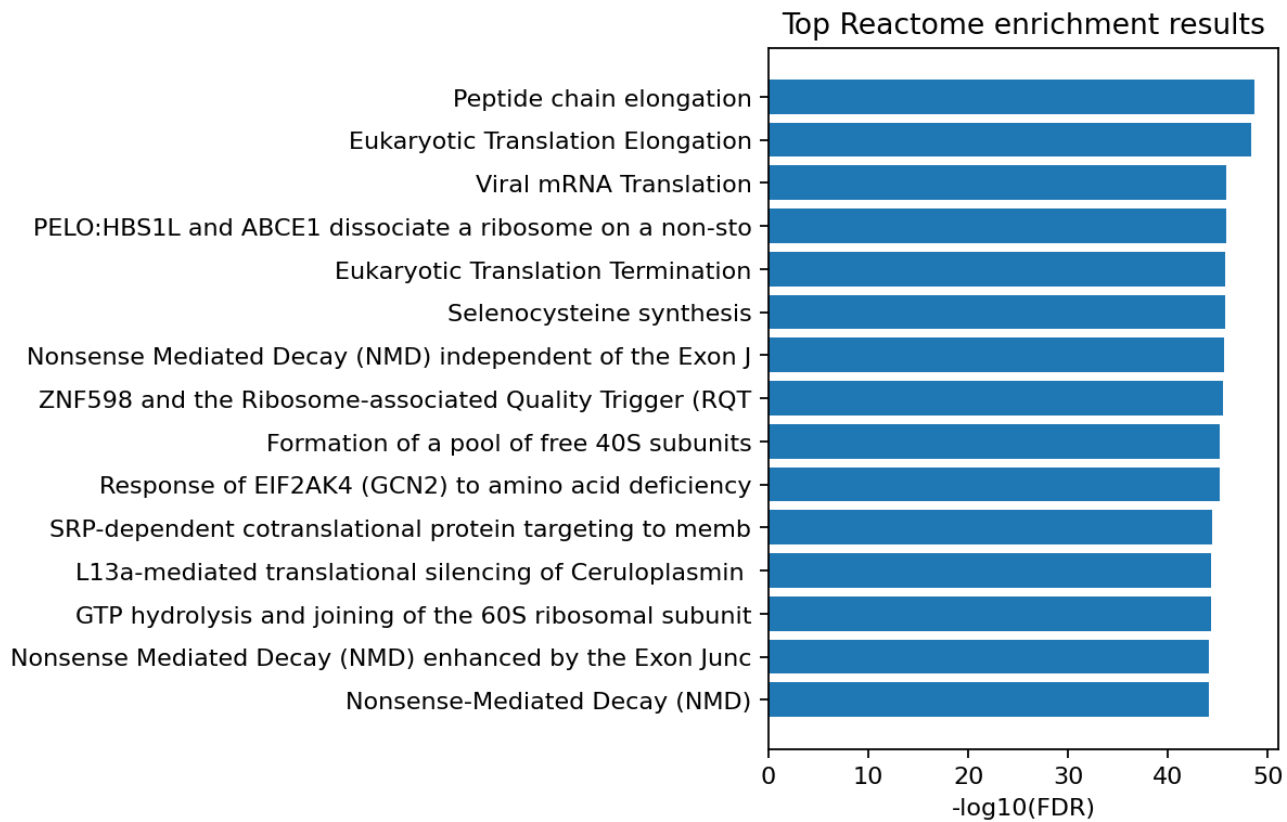


Figure 9. Most significant Reactome enrichment results across returned cell populations.

Pathway interpretation notes

- Reactome ORA is marker-set enrichment, not causal or pathway-activity proof. A non-significant pathway is not evidence that the process is absent.

Housekeeping/ribosomal audit: 159 returned top-hit row(s) have overlap genes dominated ($\geq 50\%$) by RPL/RPS genes. These hits require cautious interpretation and should not be promoted as specific tumor biology without orthogonal support.

Unassigned-population audit: 50 returned ORA row(s) come from Unassigned clusters. They are exploratory only and should not be interpreted as qualified cell-type-specific pathway findings.

Current ORA multiple-testing scope is within each cluster over pathways meeting the configured minimum-overlap gate. Treat the aggregate hit count as an enrichment-screening summary, not as a count of independent biological processes or a global pathway discovery rate.

21. Biological interpretation and scientific boundaries

The cluster annotation layer provides a cell-population framework for downstream sample-aware interpretation. Annotation remains a starting hypothesis and should be confirmed using tissue context and orthogonal evidence.

Context-specific annotation policy from the active registry profile: Automated evidence-based cell-type annotation with confidence scoring and expert-review support.

Exploratory Reactome ORA returned 995 associations passing the current within-population FDR rule. Method qualification status: pending. Aggregate counts are not a global discovery rate and do not represent independent biological processes.

Scientific interpretation boundaries

- Cells from the same biological sample are not treated as independent replicates for differential abundance or condition-level gene-expression inference.
- Differential abundance is based on sample-level cell-population compositions; pooled cell counts are not used for inferential p-values.
- CLR differential abundance is a sample-aware screening model; specialized DA methods may be preferred for confirmatory studies depending on design and biological question.
- Doublet screen is not Scrublet/DoubletFinder-equivalent.
- Ambient RNA is assessed only; no correction claim is made.
- Automated annotation is evidence-based and confidence-aware, but remains a starting hypothesis rather than definitive expert-validated ground truth.
- Populations failing the post-annotation quality gate are blocked from publishable differential-abundance interpretation without changing the underlying CLR model.

22. Scientific methods

Single-cell RNA-seq sample-aware workflow. Cell-level QC, dimensionality reduction, clustering, marker discovery and annotation are separated from biological-sample inference. Cell-population composition is summarized within each biological sample. Differential abundance starts from biological-sample compositions, then resolves the contrast-specific independent experimental unit before testing centered-log-ratio transformed compositions; paired designs use subject blocking and safely collapsible same-condition repeats use subject-level inference units. Batch terms are included when requested and identifiable; BH-FDR is applied across cell populations. Condition-level gene expression preserves raw-count biological sample $\times$ annotated cell-population pseudobulks as the aggregation artifact, then applies the same experimental-unit resolver before the selected pseudobulk differential-expression engine. The active pseudobulk DE engine was edgeR quasi-likelihood GLM. Pathway ORA used the bundled offline Reactome 2026-08-11 GMT (SHA-256 Omitted in portfolio version).

Reporting terminology contract: raw aggregation unit = sample_id; effective independent replicate unit = subject_id; blocking unit = none. Readiness and feasibility wording is rendered from the structured experimental-unit and design-identifiability status rather than recomputed from raw sample counts.

Workflow parameters

Step	Parameter	Value
qc filter	orientation	auto
qc filter	qc_mode	processed_audit
qc filter	lower_mad	3
qc filter	upper_mad	4
qc filter	mito_mad	3
qc filter	hard_max_pct_mito	30
qc filter	min_genes_floor	1
qc filter	min_counts_floor	1
qc filter	min_cells_per_gene	3
qc filter	histogram_bins	30
qc filter	scatter_max_points	2500
qc filter	min_genes_per_cell	200

qc_filter	max_genes_per_cell	7500
qc_filter	min_counts_per_cell	500
qc_filter	max_pct_mito	20
sample_metadata	cell_id_column	cell_id
sample_metadata	sample_id_column	sample_id
sample_metadata	condition_column	condition
sample_metadata	batch_column	batch
sample_metadata	subject_id_column	subject_id
doublet_detection	action	flag_only
doublet_detection	score_threshold	6
ambient_rna_correction	low_count_quantile	0.15
ambient_rna_correction	top_n	30
normalization	target_sum	10000
normalization	histogram_bins	30
hvg	n_hvg	2000
hvg	n_mean_bins	20
hvg	scatter_max_points	3000
pca	n_pcs	30
pca	random_seed	60
neighbors	n_neighbors	15
embedding	n_neighbors	15
embedding	min_dist	0.3
embedding	random_seed	60
clustering	resolution	1
clustering	random_seed	63
clustering	fallback_n_clusters	3
clustering	n_clusters	6
clustering	n_init	20
markers	marker_top_n	25
markers	max_fdr	0.05
markers	min_log_expression_delta	0.25
markers	min_pct_expressing	0.1
annotation	annotation_profile	human_pbmc_v1
annotation	species	human
annotation	biological_context	PBMC / peripheral blood
annotation	qualification_mode	Yes
annotation	reference_label_column	reference_cell_type
annotation	max_markers_per_cluster	50
differential_abundance	contrast_a	—
differential_abundance	contrast_b	—
differential_abundance	min_replicates_per_group	3
differential_abundance	zero_pseudocount	0.5
differential_abundance	include_batch	Yes
differential_abundance	include_subject_blocking	Yes
differential_abundance	top_n	100
pseudobulk	min_cells_per_pseudobulk	10
pseudobulk_de	contrast_a	—
pseudobulk_de	contrast_b	—
pseudobulk_de	min_replicates_per_group	3
pseudobulk_de	include_batch	Yes
pseudobulk_de	include_subject_blocking	Yes
pseudobulk_de	covariates	—
pseudobulk_de	top_n	200

pseudobulk_de	rscript_path	—
pseudobulk_de	timeout_seconds	300
pathway_enrichment	min_overlap	2
pathway_enrichment	top_n_per_cluster	25

23. Reproducibility and provenance

Measure	Result
Run ID	Omitted in portfolio version
Random seed	60
Workflow ID	bioflow.scrna.differential_abundance.v6.3.4
Workflow runtime version	6.3.4.3
Workflow definition SHA-256	Omitted in portfolio version
Input dataset ID	Omitted in portfolio version
Input dataset SHA-256	Omitted in portfolio version
Metadata dataset ID	Omitted in portfolio version
Metadata SHA-256	Omitted in portfolio version
LLM used	No
Network used	No

Configured qualified R / Bioconductor stack

Measure	Result
Runtime stack ID	bioflow-r45-bioc322-2026.09
Runtime baseline	bioflow.scrna.real_bioconductor_runtime.v6.3.2.1
Runtime release	6.3.2.1
R	4.5.2
Bioconductor	3.22
edgeR	4.8.2
limma	3.66.0
DESeq2	1.50.2
Runtime bundle	Omitted in portfolio version
Runtime bundle platform	darwin-x86_64
Runtime bundle SHA-256	Omitted in portfolio version
Runtime manifest fingerprint	Omitted in portfolio version

Metadata lineage

Measure						Result					
Original metadata source						single-cell_metadata.tsv					
Original metadata dataset ID						Omitted in portfolio version					
Original metadata SHA-256						Omitted in portfolio version					
Metadata scope						cell or repeated sample					
Analysis Design overlay						Yes					
Source values preserved						Yes					
Embedded into						cell_metadata, scrna_analysis_state.h5ad/obs, scrna_analysis_state.h5ad/bioflow_state/cell_metadata					
From	To	Type	Resolution Status	Confidence	Resolution Method	Automatic Confidence	Automatic Resolution Method	User Confirmed At	Mapping Fingerprint	Design Fingerprint	

scpcsa_sample_id	sample_id	semantic_canonicalization	resolved	1	user_confirmed_semantic_mapping	—	—	—	—	—
participant_id	subject_id	semantic_canonicalization	resolved	1	user_confirmed_semantic_mapping	—	—	—	—	—
scpcsa_library_id	library_id	semantic_canonicalization	resolved	1	user_confirmed_semantic_mapping	—	—	—	—	—
diagnoses	condition	analysis_design_overlay	—	—	—	—	—	—	—	Omitted in portfolio version

Metadata reconciliation

Measure	Result
Reconciliation identity role	sample_id
Reconciliation status	matched
External identity units	25
Expression-object identity units	25
Matched identity units	25
External match rate (%)	100
External-only identity units	0
Expression-only identity units	0
Conflicting mappings	0
Biological samples represented after cell join	25
Cells matched to metadata	26306
Cell coverage (%)	100
Metadata semantic resolution status	resolved
Metadata semantic fingerprint	Omitted in portfolio version
Semantic confirmation	{}

Canonical identity hierarchy audit

Measure		Result				
Cell identities		26306				
Library identities		50				
Biological sample identities		25				
Subject identities		20				
Technical replicates present		Yes				
Parent Role	Child Role	Relationship Type	Parent Units	Child Units	Max Children Per Parent	Max Parents Per Child
subject_id	sample_id	one to many	20	25	2	1
sample_id	library_id	one to many	25	50	2	1
library_id	cell_id	one to many	50	26306	1909	1
sample_id	cell_id	one to many	25	26306	3712	1

Persisted scientific-state verification

Measure	Result
Artifact	scrna_analysis_state.h5ad
Format	h5ad
Recoverable	Yes
Read-back verification passed	Yes
Persisted cells	26306
Persisted genes	31114

Cell IDs unique	Yes
Lossless cell metadata sidecar present	Yes
Metadata cells embedded into obs	26306
Metadata columns embedded into obs	93
Canonical roles embedded into obs	sample_id, subject_id, library_id, condition
Verification errors	—

Algorithms executed

Step Id	Algorithm Id	Version
qc_filter	scrna.qc.robust_mad_filter	6.3.0
sample_metadata	scrna.metadata.sample_cell_map	6.3.4.1
doublet_detection	scrna.doublet.qc_outlier	6.3.0
ambient_rna_correction	scrna.ambient_rna.low_count_assessment	6.3.0
normalization	scrna.normalization.library_size_log1p	1.1.0
hvg	scrna.hvg.dispersion_rank	1.1.0
pca	scrna.reduction.pca	1.0.0
neighbors	scrna.neighbors.knn_euclidean	1.0.0
embedding	scrna.embedding.umap	1.0.0
clustering	scrna.clustering.louvain_knn	6.3.0
cluster_de	scrna.de.welch_cluster_vs_rest	6.3.0
markers	scrna.markers.de_ranked	6.3.0
annotation	scrna.annotation.canonical_evidence	6.3.4.3.3
cell_composition	scrna.composition.sample_proportions	6.3.4
differential_abundance	scrna.differential_abundance.clr_lm	6.3.4.1
pseudobulk	scrna.pseudobulk.sum_raw_counts	6.3.1
pseudobulk_de	scrna.pseudobulk_de.edger_qlf	6.3.4.1
pathway_enrichment	scrna.pathway.reactome_oracle	6.3.0
scientific_report	scrna.report.differential_abundance_summary	6.3.4.1

24. Persisted artifact inventory

Category	Filename	Size Kib
data_export	scrna_analysis_state.h5ad	1.47129e+06
data_export	scrna_step_01_qc_filter.json	30438
data_export	scrna_step_02_sample_metadata.json	14.2
data_export	scrna_step_03_doublet_detection.json	4741.3
data_export	scrna_step_04_ambient_rna_correction.json	10.4
data_export	scrna_step_05_normalization.json	6033.3
data_export	scrna_step_06_hvg.json	13676.7
data_export	scrna_step_07_pca.json	1.1
data_export	scrna_step_08_neighbors.json	6029.9
data_export	scrna_step_09_embedding.json	14585.7
data_export	scrna_step_10_clustering.json	2.1
data_export	scrna_step_11_cluster_de.json	402966
data_export	scrna_step_12_markers.json	359.8
data_export	scrna_step_13_annotation.json	260.1
data_export	scrna_step_14_cell_composition.json	66.6
data_export	scrna_step_15_differential_abundance.json	11.9
data_export	scrna_step_16_pseudobulk.json	29.8
data_export	scrna_step_17_pseudobulk_de.json	18.2
data_export	scrna_step_18_pathway_enrichment.json	216.4
data_export	scrna_step_19_scientific_report.json	65.8

table	scrna_cell_embedding.csv	5627.6
table	scrna_marker_genes.csv	122.1
table	scrna_cluster_annotations.csv	13.8
table	scrna_annotation_evidence_coverage.csv	4.6
table	scrna_annotation_assignment_audit.csv	7.8
table	scrna_annotation_candidate_evidence.csv	42.9
table	scrna_annotation_condition_stability.csv	0.5
table	scrna_annotation_quality_gate.csv	1.1
table	scrna_hvg.csv	4611.6
table	scrna_hvg_audit.csv	4611.6
table	scrna_normalization_cell_audit.csv	2213.9
table	scrna_qc_cell_metrics.csv	1825.4
table	scrna_qc_filter_audit.csv	2669
table	scrna_qc_gene_filter_audit.csv	4355.2
table	scrna_doublet_cell_audit.csv	1482.3
table	scrna_ambient_rna_risk_genes.csv	3.2
table	scrna_cluster_differential_expression.csv	143874
table	scrna_sample_qc_summary.csv	2.3
table	scrna_sample_design.csv	0.7
table	scrna_condition_sample_counts.csv	0
table	scrna_cell_composition_condition_summary.csv	1.6
table	scrna_pseudobulk_group_summary.csv	6.3
table	scrna_pseudobulk_de_readiness.csv	2.1
table	scrna_inference_semantics.csv	0.8
table	scrna_reactome_enrichment.csv	79.9
table	scrna_cell_composition_by_sample.csv	15.5
table	scrna_pseudobulk_raw_counts.csv	32858.6
report	scrna_scientific_report.json	62.2
report	scrna_methods.txt	1.5
report	scrna_scientific_report.html	4.7
figure	scrna_qc_distributions.png	38.6
figure	scrna_pca_explained_variance.png	34.5
figure	scrna_embedding_cluster.png	88.8
figure	scrna_embedding_annotation.png	95.5
figure	scrna_cluster_sizes.png	26.9
figure	scrna_cell_composition.png	54.1
figure	scrna_reactome_top_pathways.png	102.4
report	scrna_supplementary_results.xlsx	40865.5

To keep the main report readable, long storage paths and download URLs are not repeated here. Exact relative paths and the complete artifact manifest remain available in scRNA Supplementary Results.xlsx, the Full Results Package, and the analysis download interface.

Primary delivery files: scrna_scientific_report.docx (main report), scrna_supplementary_results.xlsx (detailed review tables), and scrna_full_results_package.zip (complete lossless artifacts). Representative source tables include scrna_qc_cell_metrics.csv, scrna_cluster_differential_expression.csv, scrna_cell_composition_by_sample.csv, scrna_differential_abundance.csv, scrna_pseudobulk_condition_de.csv, and scrna_reactome_enrichment.csv.

Report completeness note

Completeness is provided across three coordinated layers rather than by copying every row into Word. The main report contains the scientific narrative, summary statistics, figures, full sample/annotation tables, and ranked representative results. The Supplementary Results workbook contains detailed review tables. The Full Results

Package preserves the complete persisted CSV/JSON outputs, figures, methods, provenance, and report files without loss.